# Undertale: A Case Study in Ludomusicology

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Approved by _____

Date _____

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*Soli Deo Gloria*

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## *Introduction: An Interactive Experience*

"In this world, it's kill or be killed."

Gamers are too often faced with this ultimatum when playing video games, but what happens when this paradigm is turned on its head? How do psychology, music, and choice play into the profound mess that is ethics in gaming? My thesis is a consideration of these two questions in light of the impact which *Undertale* (an independently developed title) has had on recent gaming history. In short, my writing is an in-depth analysis of how music is used in this title to reflect the developer's conception of morality, demonstrating how tonal music is used to reward a player's "good" choices while tonal ambiguity is used to punish game participants who take part in "evil" behavior. Simultaneously, I discuss the soundtrack's effectiveness, pointing out a variety of compositional techniques used to complement the game's narrative. These observations ultimately take the form of a critique of the overall gameplay experience and presentation of in-game ethics.

The game, designed by Toby Fox, was largely shaped by tropes from internet culture and the developer's personal experiences with video games. Fox wrote the script, created the overall design, and composed the soundtrack for his work while receiving additional artwork from Temmie Chang. Released in September of 2015, *Undertale* was well received and was quickly given a "cult classic" status by multiple publications. This raises the question of what about this game is so appealing to gamers and serious game critics alike, and the answer is simple: catchy music, strong narrative, and memorable characters. *Undertale* has a story that deals with the deeper issues of our humanity, and its plot is only further enhanced by Fox's ability to compose music capable of affecting its listeners. Research shows that "gameplay, as a human experience, is instilled with emotions", so it stands to reason that a game which appeals to a player's empathy

would have a strong impact on the gaming community.[1] As for the in-game characters, each of them has a unique personality and goes through significant developments of their own over the course of the narrative's unfolding.

Players interact with the game from the perspective of a young child named Frisk who has been trapped in the Underground (a fantastical setting inhabited by a thriving community of monsters). During the gaming experience, the character quickly learns that the environment is hostile and that the cave dwellers believe that the power contained within a human soul is the only thing that can set them free from their subterranean imprisonment. Thus, the player is given two choices: they either can hack and slash their way through the monsters in order to find their way out of the caves, or they can do something different, something unprecedented for this genre of gameplay. The other option presented to the player is that of pacifism. If it so desired, the narrative presentation in *Undertale* can be entirely free of player-initiated violence, yielding control of Frisk's characterization to the game participant. Due to the inclusion of this "choose your own adventure" component in the interactive experience, the game's story can come to a close with a variety of ending sequences.

*Undertale* uses music to strengthen its commentary on ethics in gaming, a method which ultimately affects its players' perception of violence, sound, and choice. This claim is important not only because of the work's potential to become a landmark title in video game history, but also due to the fact that no one has yet published any academic scholarship on either *Undertale* or the correlation between soundtracks and their effect on one's conception of morality. My research shows how Fox designed his music to be capable of not only complementing his narrative, but also influencing players in regards to their decisions throughout the gaming

1 Aki Järvinen, "Understanding Video Games as Emotional Experiences," in *The Video Game Theory Reader 2*, ed. Bernard Perron and Mark Wolf (New York: Routledge, 2009), 86.

experience. In essence, I am proposing that *Undertale* does much more than the average game does. It is a work which "succeeds where a number of games that promise 'moral choice' fail: by presenting the player with true choices, merging those choices into its gameplay and narrative, and being consistent and firm about the consequences of those choices."[2] Though I later discuss how the title's presentation of ethics is confusing at best, *Undertale* truly affords its players a world in which their decisions have lasting repercussions. As the field of ludomusicology continues to grow, scholars will need to begin looking at video game music from more perspectives than they previously have been, and my research can provide some new questions for academics from various fields to consider.

The way I went about my study was more or less simple: I played *Undertale* for myself, catalogued the choices given to me, compared my findings to those of game critics, and then analyzed these items in regards to how they interact with the soundtrack. When I needed to revisit the game for the investigation of a certain event, I turned to YouTube videos of gameplay footage in order to review character dialogue, player choices, and musical cues. As for which choices were given the most attention in my analysis, I decided to spend more time talking about the decisions which significantly affect the interactive experience. However, less important choices are also discussed provided that I found them unique to *Undertale*'s overall presentation. By then comparing my research with that of online game reviews, I was able to contextualize my findings within the gaming community itself. This was helpful to me since gamers are passionate when writing about the games they do and do not enjoy, thus providing me with an abundance of data in terms of raw, emotional reactions to the ethical dilemma this game portrays. The final step of my methodology involved an analysis of the soundtrack's various musical affects and

2 "[Fight] or [Mercy]," last modified October 19, 2015, http://geekandsundry.com/fight-or-mercy-challenging-player-morality-in-undertale.

how they complement both the narrative and choices given to the player. Each major scene in which a decision must be made makes use of a particular track in order to heighten its dramatic tension, and I have closely examined these elements of choice and music as a way of revealing how the two interact.

The study of video game music is a recent topic of scholarship explored in books, articles, and dissertations written by both academics and amateurs. Though there are articles that date as far back as 1987 on the subject of music technology in gaming, commentary on its artistic value and effect on culture would not surface until the late 1990s. Examples from this decade include Jessie Cameron Herz's *Joystick Nation: How Videogames Gobbled Our Money, Won Our Hearts, and Rewired Our Minds*[3] (1997) and Matthew Belinkie's "Video Game Music: Not Just Kid Stuff "[4] (1999). Although these writings are not strictly academic in nature, they would open the floodgates for an outpouring of articles and theses on the topic of video game music in the early 2000s. One of the first, dedicated musicological books to emerge with the aim of tackling the subject of game soundtracks (several publications dealing with the practical aspects of music technology and the game scoring industry were already written by this time) was Karen Collins' *Game Sound: An Introduction to the History, Theory, and Practice of Video Game Music and Sound Design*[5] (2008). This book, and others like it, would pave the way for the birth of an "inter-university … organization dedicated to the study of game music" known as the Ludomusicology Research Group in August of 2011.[6] Since then, numerous books and articles

3 Jessie Cameron Herz, *Joystick Nation: How Videogames Gobbled Our Money, Won Our Hearts, and Rewired Our Minds* (London: Abacus, 1997), 1-230.

4 "Video Game Music: Not Just Kid Stuff," last modified December 15, 1999, http://www.vgmusic.com/information/vgpaper.html.

5 Karen Collins, *Game Sound: An Introduction to the History, Theory, and Practice of Video Game Music and Sound Design* (Cambridge: MIT Press, 2008), 1-216.

6 "About," Ludomusicology Research Group, accessed September 30, 2016, http://www.ludomusicology.org/about.

have been released under the banner of ludomusicology including *Music and Game: Perspectives on a Popular Alliance*[7] (edited by Peter Moormann in 2013), William Cheng's *Sound Play: Video Games and the Musical Imagination*[8] (2014), and *Music Video Games: Performance, Politics, and Play*[9] (edited by Michael Austin in 2016).

The perception of video games as they relate to psychology and player interaction has changed over the years. In "Videology: Video-Games as Postmodern Sites/Sights of Ideological Reproduction"[10] (1995), Simon Gottschalk suggested that all video games are organized around a particular set of ideologies. The conclusion he came to was that this particular form of interactive entertainment revolves around the concepts of violence, "othering", individualism, the helplessness and sexualization of women, the rejection of drug use, commercialism, over stimulation, and fixed rules. In his writing, the author related these ideas to the then-current movement toward a postmodern culture. John Seel (in the same spirit as Gottschalk) would later argue how technology and interactive entertainment are embedded in youth culture and how they have the potential to influence character development in his article "Plugged in, Spaced out, and Turned on: Electronic Entertainment and Moral Mindfields"[11] (1997). This author suggested that gaming for younger children could be particularly formative due to their inability "to make the associative distinctions between fantasy and reality."[12] Seel further proposed that video games detach their players from a sense of authentic perception, thus complicating their ability to form

7 Peter Moormann, ed., *Music and Game: Perspectives on a Popular Alliance* (Wiesbaden: Springer VS, 2013), 1-223.

8 William Cheng, *Sound Play: Video Games and the Musical Imagination* (Oxford: Oxford University Press, 2014), 1-272.

9 Michael Austin, ed., *Music Video Games: Performance, Politics, and Play* (New York: Bloomsbury Academic, 2016), 1-352.

10 Simon Gottschalk, "Videology: Video-Games as Postmodern Sites/Sights of Ideological Reproduction," *Symbolic Interaction* 18 (1995): 1-18.

11 John Seel, "Plugged in, Spaced out, and Turned on: Electronic Entertainment and Moral Mindfields," *The Journal of Education* 179 (1997): 17-32.

12 Seel, "Plugged in, Spaced out," 22.

character based on real-world situations. Years later, scholars would need to address the predictions which these and other authors had made. In response to the political controversy surrounding gaming and the legality of distributing adult-oriented content, Joel E. Collier, Pearson Liddell Jr., and Gloria J. Liddell commented on (in their 2008 article "Exposure of Violent Video Games to Children and Public Policy Implications"[13]) how United States legislators find it difficult to pass laws in favor of restricting minors from playing certain video games. They continued by asserting that quantifiable data and clearly worded proposals are what needed to be presented in order to pass new legislature regarding the limiting of children's access to games containing violent or sexual themes, and that the previously discussed theories regarding gaming and moral development in minors have yet to be validated.

Andreas Gregersen and Torben Grodal would neglect the conversations surrounding violence in favor of examining how players physically and mentally interact with virtual environments through their article "Embodiment and Interface"[14] from *The Video Game Theory Reader 2*. Their research suggests that video game controllers offer their users varying degrees of extended embodiment, and the study of these occurrences has opened the door to further discussion on an assortment of topics such as the empathetic relationship between players and their virtual avatars. Later, in an effort to understand why people choose to play video games, Bobby Hoffman and Louis Nadelson would carry out a study ("Motivational engagement and video gaming: a mixed methods study"[15]) in which they sought to discover the motivation behind gaming. While the most frequently cited reason for playing video games was to escape from

13 Joel E. Collier, Pearson Liddell Jr., and Gloria J. Liddell, "Exposure of Violent Video Games to Children and Public Policy Implications," *Journal of Public Policy & Marketing* 27 (2008): 107-112.

14 Andreas Gregersen and Torben Grodal, "Embodiment and Interface," in *The Video Game Theory Reader 2*, ed. Bernard Perron and Mark J.P. Wolf (New York: Routledge, 2009), 65-84.

15 Bobby Hoffman and Louis Nadelson, "Motivational engagement and video gaming: a mixed methods study," *Educational Technology Research and Development* 58 (2010): 245-270.

reality, the authors also listed stress relief and a perceived sense of accomplishment as important factors. Furthermore, their research implies that the ability to control one's gaming environment and the social aspect of gaming culture are what draw many to engage in frequent gameplay sessions. Patrick Osborne would then revisit the earliest approaches to understanding the psychology of gaming by asking the question of why virtual violence was so commonly present in video games through his article "Evaluating the Presence of Social Strain in Rockstar Games' 'Grand Theft Auto IV'"[16] (2011). By analyzing the 2008 release Grand Theft Auto IV, the author proposed that the title's narrative "encourages lawlessness rather than encouraging the player's conformity to societal norms."[17] Osborne further suggested that the social phenomena and ideologies being expressed to potentially impressionable players are more influential than the game's violent imagery. The author's writing concludes by demonstrating the various ways in which this title teaches its players how societal strain is a justifiable cause of the protagonist's violent behavior and rebellious lifestyle.

In regards to video game "performativity", my research includes both scholarly articles on the idea of gaming itself as well as writings which deal with the subject of popular YouTube performers. By giving a brief overview of YouTube's history and use in an academic setting, Ayanna Thompson was able to bring to light the platform's potential as an educational tool in her article "Unmooring the Moor: Researching and Teaching on You Tube"[18] (2010). Moreover, the author's research suggests that YouTube is a social construct centered on the ideas of dialogue and response. Though many of the website's users engage in passive viewing, it is the

16 Patrick Osborne, "Evaluating the Presence of Social Strain in Rockstar Games' 'Grand TheftAuto IV'," *Studies in Popular Culture* 34 (2011): 109-132.
17 Osborne, "Evaluating the Presence of Social Strain," 111.
18 Ayanna Thompson, "Unmooring the Moor: Researching and Teaching on You Tube," *Shakespeare Quarterly* 61 (2010): 337-356.

interactions between YouTube performers and their fans that have kept the network growing. Christine B. Balance (in her 2012 article "How It Feels to Be Viral Me: Affective Labor and Asian American YouTube Performance"[19]) would then cover a variety of topics from "YouTuber" Jimmy Wong's viral video "Ching Chong! Asians in the Library Song" to the factors which affect a YouTube video's popularity. The author proposed that in order for a video to attract attention it must contain an "emotional hook", function as a form of affective labor, and display a certain level of both authenticity and sincerity. While measuring YouTube's popularity and growth, Gary Weidenaar presented a number of statistics with the intention of informing both educators and musicians of the benefits inherent to using social media as a means of promotion through his article "Facebook and YouTube and Choirs – Oh My!"[20] (2013). His then-current measurements provide insight into the way these platforms have grown over the past few years. Furthermore, the author's work suggests that YouTube is making the transformation from a simple warehouse for videos and digital information to a performance-oriented network capable of garnering the attention of business professionals. In regards to gaming, Michael Liebe would directly approach the subject of video game performativity by both discussing the circumstances necessary for an interactive experience and postulating that gameplay has become a genre of performance in his "Interactivity and Music in Computer Games"[21] from *Music and Game: Perspectives on a Popular Alliance* (2013). The author begins with an exploration of "interactivity", asserting that games primarily rely on human interaction whether it takes place between an individual and the game itself or any number of non-computer players. Following

19 Christine B. Balance, "How It Feels to Be Viral Me: Affective Labor and Asian American YouTube Performance," *Women's Studies Quarterly* 40 (2012): 138-152.
20 Gary Weidenaar, "Facebook and YouTube and Choirs – Oh My!," *The Choral Journal* 10 (2013): 57-60.
21 Michael Liebe, "Interactivity and Music in Computer Games," in *Music and Game: Perspectives on a Popular Alliance*, ed. Peter Moorman (Wiesbaden: Springer VS, 2013), 41-62.

this is Liebe's discussion of music games as they relate to performativity in general, live events, and the dedicated communities they have generated.

When compared to the fields of opera or musical theater, there has been little scholarly research in regards to how music relates to in-game narrative. While there are plenty of articles discussing what makes game music interesting, and a number of masters' theses which discuss the intersection between story and sound, I found that studying these works in conjunction with articles on the topic of film and music was exceedingly beneficial since ludomusicology is often influenced by scholarship on movie scoring. The theories presented in these writings, equally applicable in the world of interactive entertainment, have developed significantly over the years. In Claudia Gorbman's "Narrative Film Music"[22] (1980), a clear distinction is made between diegetic and non-diegetic music in film, demonstrating how an understanding of these differentiations can be used to express a wide variety of ideas to the audience. The author further suggested that our approach to both analyzing and critiquing scores needs to be based on entirely different standards than the ones used for absolute music. Because of the interaction between soundtrack and narrative, Gorbman proposed that our criticism of dramatic scoring be informed by an understanding of its purpose. James Lendino would later discuss some of the most basic principals of composing video game soundtracks, asserting that music needs to complement gameplay.[23] Although this concept comes across as an obvious necessity to modern game developers, it was an idea which had yet to take root in less successful gaming companies. The author further spoke in favor of soundtracks which are able to smoothly transition from ambient music to strong, melodic themes, thereby drawing its listeners into a title's environment and

22 Claudia Gorbman, "Narrative Film Music," *Yale French Studies* 60 (1980): 183-203.

23 James Lendino, "Scoring for the Modern Computer Game" (paper presented at the International Computer Music Conference, San Francisco, California, 1998).

narrative arc. Arguing that film scores are an essential part of the storytelling experience by communicating various types of information to viewers, Jessica Green added another layer of depth to the analysis of film music in her article "Understanding the Score: Film Music Communicating to and Influencing the Audience"[24] (2010). Her research unveiled just how complex the use of scoring could be, and divided incidences of film music into the following categories: redundant, contrapuntal, empathetic, a-empathetic, and didactic contrapuntal. Though not necessarily contrary to Gorbman's research, Green's findings imply that soundtracks utilize connections between themes, motivic transformation, leitmotifs, and other such techniques as complex as those found in operas and musicals. Around the same time, Pieter Crathorne (in his 2010 thesis "Video game genres and their music"[25]) covered a variety of topics by analyzing the music in titles from a wide array of video game genres. Throughout his discussion, the author proposed that thematic materials are often associated with both characters and gaming environments, and that these connections can enrich a player's experience. Crathorne also commented on the use of real-world music for symbolic purposes, a technique which can influence one's perception of a title's dramatic arc.

Motivic structures play an important role in the presentation of musical narrative, and have been analyzed from the perspectives of both perception and theming. In Jason Brame's relatively recent analysis of soundtracks from The Legend of Zelda franchise, motivic and "prolongational" analyses were used to suggest a thematic unity between games in the series.[26] By reducing various themes to their structural levels, Brame has proposed a kind of motivic

24 Jessica Green, "Understanding the Score: Film Music Communicating to and Influencing the Audience," *The Journal of Aesthetic Education* 44 (2010): 81-94.

25 Pieter Crathorne, "Video game genres and their music" (master's thes., University of Stellenbosch, 2010).

26 Jason Brame, "Thematic Unity Across a Video Game Series," *ACT - Zeitschrift für Musik & Performance* 2 (2011): 1-16.

narrative by comparing the formal and intervalic qualities from corresponding themes in the franchise. Axel Stockburger would then further clarify the distinctions between the various sound objects found in soundtracks by clearly defining what diegetic and non-diegetic game music is through his paper "The Game Environment from an Auditive Perspective"[27] (2013). His assertion is that both diegetic and non-diegetic sounds permeate gaming environments as a means of creating a dynamic virtual world. Due to the noted interaction of sound and gameplay, the author concluded that these sound objects are not only user driven, but also an essential part of the narrative experience. More recently, Michiel Kamp (in his 2014 article "Musical Ecologies in Video Games"[28]) sought to find the link between player interaction and in-game musical cues by examining the relationship between soundtracks and virtual narrative. The author's writing includes a variety of case studies in which the implications of musical gestures as they relate to a game's rules and boundaries are discussed. Ultimately, Kamp came to the conclusion that video game music can affect a player's emotional state and understanding of perception when seriously engaged in gameplay. In regards to scholarship pertaining to choice and how this affects a player's gaming experience, Mark J.P. Wolf wrote his article "Assessing Interactivity in Video Game Design"[29] in 2006 on the topic of interactivity in video games. In his writing, the author discusses the concept of a game's "replayability" as it relates to the number of choices a player is presented with. Wolf's claim is that the more choices a game offers its users, the more likely someone will replay it in order to explore the consequences of taking different actions. Furthermore, the author uses choice as a way of measuring a game's level of interactivity, suggesting that the motivation behind these choices needs to be considered when

27 Axel Stockburger, "The Game Environment from an Auditive Perspective" (paper presented at the Digital Games Research Association, Atlanta, Georgia, 2013).
28 Michiel Kamp, "Musical Ecologies in Video Games," *Philosophy & Technology* 27 (2014): 235-249.
29 Mark J. P. Wolf, "Assessing Interactivity in Video Game Design," *Mechademia* 1 (2006): 78-85.

discussing player interaction.

Since there have yet to be any scholarly writings about *Undertale* in particular, my only option in regards to contextualizing my observations was to compare them with those of leading authorities in the world of gaming. While some mainstream websites offer unbiased reviews, entries from other such sources are clearly opinion pieces. However, both of these writings have helped me to frame my arguments and conclusions while gaining an understanding of players' general perception of the gameplay experience. Articles have been drawn from video game and "nerd culture" news websites like Geek and Sundry, Kill Screen, and PC Gamer with titles such as "[Fight] or [Mercy]? Challenging Player Morality in Undertale"[30], "Undertale's Not as Peaceful as it Pretends to Be"[31], and "It's OK if You Killed Everyone in Undertale"[32]. Interestingly enough, nearly every author has discussed the fact that the choices made in *Undertale* have serious, plot-altering consequences. Other observations include the consistency with which morality is dealt with in Fox's narrative, the high level of player interactivity in regards to how users' actions shape the gameplay experience, and the way in which the game's soundtrack complements its themes and the development of the title's characters.

In summary, my writing combines a variety of aspects from the aforementioned research to thoroughly examine *Undertale* with the intention of explaining its complex combination of psychological underpinnings, musical scoring, and choice-driven gameplay. In chapter one, I will go into further detail regarding gaming and psychology, interactivity, performativity, soundtrack composition, and music as narrative in order to prepare the reader for my case study

30 "[Fight] or [Mercy]."
31 "Undertale's Not as Peaceful as it Pretends to Be," Kill Screen, accessed May 4, 2016, https://killscreen.com/articles/undertales-not-peaceful-it-pretends.
32 "It's OK if You Killed Everyone in Undertale," last modified November 5, 2015, http://blog.plasterbrain.com/2015/11/05/genocide.

of the game in question. Each of these topics will be explored through the lens of modern scholarship as I set the stage for an exploration of how *Undertale* uses its various features to present its players with a unique, ethically charged experience. The second chapter of my thesis will serve as the beginning of my investigation by carefully surveying the game's use of motives to influence a player's perception of both narrative and the title's characters. This analysis includes commentary on the game's psychological aspects, soundtrack, presentation of choice in gameplay, and how these features work together in order to condition players to behave a certain way. The final chapter of my work will explore *Undertale*'s use of tonality in regards to its system of in-game ethics. By including an analysis of how the music, dialogue, and number of available choices change as a player makes their decisions, this chapter will show how Fox's narrative is supported by every facet of the game's internal structure.

## *Chapter One: On Gaming, Performance, and Soundtracks*

John Seel has claimed that children involved in gaming are "no longer a passive spectator, but an active participant in a rule-driven microworld of frenetic action."[1] And indeed, games do shape the way both children and adults think about the world around them. In Gerard Jones' *Killing Monsters: Why Children Need Fantasy, Super Heroes, and Make Believe Violence*, the author argues that "violent video games provide a safe fantasy world in which children can learn to control the emotions of anger, violence, and sexuality."[2] Others have made attempts at linking aggressive behavior to violent video game exposure, although no such association has been validated by scholarly research.[3] Ultimately, there is only one thing we can be certain about: gaming produces a response within gamers which is comparable to the reactions incited by listening to music or watching theater productions. Moreover, video games provide the intuitive, immediate feedback one experiences while playing a sport or engaging in other physical activities. Because of this duality, games have the ability to teach their players anything from world history to the finer points of pixelated ghost evasion. Just as there is a contrast between radio jingles and symphonies, the quality of video game content can range from cheap and mindless to the artistic and profound. But in order to make sense of it all, I will discuss a variety of topics from the motivation behind gaming to video game music as narrative while giving examples from popular titles. This will give readers an understanding of the analytical techniques and perspectives I have utilized in my analysis of *Undertale*. But before I present these concepts, I will summarize the title's plot in order to frame my arguments.

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1 John Seel, "Plugged in, Spaced out, and Turned on: Electronic Entertainment and Moral Mindfields," *The Journal of Education* 179 (1997): 21.

2 Joel E. Collier, Pearson Liddell Jr., and Gloria J. Liddell, "Exposure of Violent Video Games to Children and Public Policy Implications," *Journal of Public Policy & Marketing* 27 (2008): 109.

3 Collier et al., "Exposure of Violent Video Games," 109.

## I. Plot Summary

*Undertale*'s in-game history (that is, the record of events prior to the protagonist's introduction) begins with humans and monsters living in harmony, coexisting on the Earth's surface. However, a war eventually breaks out between the two nations due to the humans' fear of the monsters' ability to absorb souls, leading to the defeat and imprisonment of the monster race. Now banished to the Underground, the monsters become resentful toward the humans living above them. Sometime later, a human child named Chara falls into the caves and is adopted by Asgore and Toriel, the king and queen of the Underground. Raised alongside Asriel (the royal prince), Chara is treated as if they belong to the family. However, the human child has malevolent tendencies and intends on escaping to the Earth's surface for the purpose of wreaking havoc on humanity. Chara's plan is to commit suicide, have Asriel absorb their soul, and then kill as many people as possible in order to grow stronger through the absorption of human souls. Though the royal prince is tricked into taking part in the plan, he is able to stop the human child from completing their malicious scheme. Resulting in the death of both Chara and Asriel, this series of events is what motivates Asgore to declare a second war against humanity, killing any humans who happen to fall into the Underground.

After an unspecified number of years pass, another human child by the name of Frisk falls into the caves. Players are given complete control of this character, taking the role of the title's main protagonist. In short, Frisk's goal is to escape the Underground in order to return to the Earth's surface while dealing with number of unique characters along the way. For players who choose to avoid violent behavior, they are able to secure a path to freedom for the child while developing relationships with the subterranean monsters. In this case, they must ultimately face Flowey (a sentient flower containing the soul of Asriel) in an attempt to free themselves as

well as those already trapped beneath the Earth. If successful, both Frisk and the entirety of the monster nation are freed from their underground imprisonment, resulting in a joyful ending. As for players who choose to eliminate each of the monsters residing in the caves, they are punished with notable changes to the title's soundtrack, permanent alterations to the gameplay experience (explained in further detail later on) and a sequence in which Chara is resurrected. Upon reaching this scene, the player is met with a morbid lecture as delivered by the murderous child. In order for a game participant to restart the game, whether to rectify their actions or otherwise, they must agree to give Chara their soul.

The way in which players are given the power to influence the unfolding of *Undertale*'s narrative speaks to Fox's ability as a creative game developer. The techniques used (in terms of both the title's soundtrack and interactive experience) while crafting his work allowed for it to quickly grow in popularity amongst gamers. But before we examine the driving forces behind the success of this independently developed title, it is necessary to first understand some of the basic principals surrounding gaming, performance, and soundtracks. This will give my case study of the game in question a framework in which to be analyzed. To begin, we will start with some scholarly research on the topic of player psychology.

## II. The Motivation for Gaming

In a study carried out by Bobby Hoffman and Louis Nadelson, their subjects' "most frequent reason cited for engaging in game play was to escape from daily routines such as school or work."[4] It seems that many people play video games because they have the desire to find another reality in which to exist and play. But while escapism is certainly a factor in regards to why games have grown in popularity, there are a number of other reasons which motivate

4 Bobby Hoffman and Louis Nadelson, "Motivational engagement and video gaming: a mixed methods study," *Educational Technology Research and Development* 58 (2010): 257.

players to devote their time to gaming. Not surprisingly, participants of the aforementioned study expressed that "gaming was a diversion to relieve stress and in many cases fostered feelings of well-being and accomplishment."[5] I would like to suggest that the impetus behind this emotional response can be attributed to gaming "achievements". These achievements (sets of in-game goals ranging from typical gameplay milestones to nearly impossible challenges) are awarded to players by either in-game reward systems or their console's online infrastructure such as the PlayStation Network or Xbox Live. The completion of such challenges makes players feel good since it is an indication of their skill, intellect, or perseverance. Furthermore, achievements are displayed for all to see, so it stands to reason that gamers would find a sense of accomplishment and pride in fulfilling these sets of tasks from a game or game series.

Hoffmann and Nadelson's findings also suggest that a player's "ability to control their gaming characters, the gaming environment, such as handheld controls, and the degree of the challenge, were important motivations to play a video game."[6] Since 2010, the year in which the authors' study was carried out, video games have become more choice-driven than ever before, and players now have access to games in which they are afforded the ability to play in whatever manner they most prefer. A great example of this is the August 2016 release *No Man's Sky*, an action-adventure survival game which is set in a procedurally generated universe capable of producing eighteen quintillion unique planets which players can visit at their leisure (Ex. 1.1). This over-the-top allowance of choices puts that control (and perhaps too much of it) into the hands of the player. Lastly, the study strongly implied that "playing with others was a significant predictor of motivation to play video games."[7] Humans are social creatures, and it is no surprise

5 Hoffmann and Nadelson, "Motivational engagement," 257.
6 Hoffmann and Nadelson, "Motivational engagement," 262.
7 Hoffmann and Nadelson, "Motivational engagement," 262.

*Example 1.1. No Man's Sky, Hello Games, landing on an uncharted planet*




that socialization has become an increasingly prominent factor in regards to why gaming is becoming more popular. Multiplayer features abound in most modern video games, and players cry out when game developers fail to deliver them. *Pokémon Go*, released for mobile platforms in July of 2016, initially featured a system in which players could capture a variety of trademarked creatures, join a faction, and have their newly caught Pokémon battle with other monsters left by players of opposing factions at specific "Pokémon gym" locations. However, direct player-to-player interaction was not an included feature upon the application's launch (much to the disappointment of fans of the franchise), and the game's developers have since promised to release updates incorporating social components in the near future.

## III. Player Interactivity

John Seel has suggested that video games were "the first medium to combine visual dynamism with active participation".[8] For this reason alone, we can say that games are unique in their ability to combine a variety of art forms with elements of interactivity. Furthermore, this is important to note since human interaction is the primary impetus on which most games operate. A good example of an exception to this observation are simulation games, which rely heavily on their reactions to randomly generated, in-game circumstances. That aside, some of the earliest video games (such as *Pong* or *Space Invaders*) relied exclusively on player-controlled inputs, prioritizing human interaction above all else. But interactivity has come a long way since the early 1970s, and the notions of participation have changed quite a bit in this short period of time. In fact, entrepreneur and gaming consultant Michael Liebe has commented on how games themselves have the ability to "generate spaces (realistic or non-realistic) to be explored by the players."[9] Gaming environments (instead of being meticulously laid out by game developers who have specific landscapes in mind) can now be procedurally generated through the miracle of computer programming. Though neither approach to creating an in-game world is necessarily superior, they both allow players to traverse and interact with vast, dynamic spaces which explore the depths of human creativity. *Morrowind*, originally released for personal computers in May 2002, features a game world approximately sixteen square kilometers in size. Moreover, the title features hundreds of unique locations for players to explore, various fictitious cultures to interact with, and a rich history as recorded in nearly three hundred readable, in-game books. Clearly, the possibilities for interaction being offered to players is at an all time high.

8 Seel, "Plugged in, Spaced out, and Turned on," 22.

9 Michael Liebe, "Interactivity and Music in Computer Games," in *Music and Game: Perspectives on a Popular Alliance*, ed. Peter Moorman (Wiesbaden: Springer VS, 2013), 46.

Liebe further asserts that all games "rely on interaction – be it with one's partner, or one's opponent or with the game system itself".[10] Although the degree to which this interaction actually affects a user's experience varies from title to title, we cannot deny the significance that participation plays in regards to a game's purpose. A game of chess where your opponent refuses to make a move may as well be renamed "conscientious objector", and a computer game left unplayed is as useless as the unavailable digital space it occupies on your hard drive. In a way, a game is not really a game until it is played.

Consider the *Call of Duty* franchise which boasts some of the most realistic computer and gaming console graphics in the video game industry. Because of the series' immersive, real-world sounds and visuals, it is sensible for scholars such as Andreas Gregersen and Torben Grodal to suggest that "interacting with video games may lead to a sense of extended embodiment"[11] After all, a game which successfully engages a player's senses is one that is easy to become immersed in. This is particularly true in regards to first-person survival games in which the protagonist is met with a regular bombardment of auditory and visual cues. In this genre, music is used to heighten the senses at particular moments, visuals can cause a player to visibly flinch or even jump out of their seat, and gameplay demands that users are fully engaged at all times. Moreover, these elements play an important role when it comes to the concept of a player's sense of embodiment.

It has been argued that video game controllers, which come in a variety of shapes and sizes depending on the console being utilized, "provide a mapping functionality that allows us to perform a wide range of actions in relation to that game system and its virtual environment."[12]

10 Liebe, "Interactivity and Music," 49.
11 Andreas Gregersen and Torben Grodal, "Embodiment and Interface," in *The Video Game Theory Reader 2*, ed. Bernard Perron and Mark J.P. Wolf (New York: Routledge, 2009), 67.
12 Gregersen and Grodal, "Embodiment and Interface," 69.

This functionality draws players from their sense of self into a receptive state in which they can empathize with their virtual avatar. Though manipulating controllers is a far cry from the feeling of actually handling a broadsword or driving a race car, players who become immersed in gaming lose track of such thoughts and begin relying on their immediate responses to a game's stimuli. The degree to which this phenomenon occurs largely depends on a number of factors from sound design to visual elements, but is perhaps most strongly influenced by gameplay and how much a player relates to their avatar. Indeed, a game that engages its user's attention and is capable of instilling a sense of empathy for its protagonist will affect the psychological state of a player, mentally immersing them into another reality. Games developed by teams who have neglected (whether intentionally or not) to include such design choices are less engrossing, though not necessarily less entertaining. In either case, this immersion is what Gregerson and Grodal are suggesting by using the term "extended embodiment", but their research did not stop there. The authors went further by proposing that motion controlled interfaces (which require motions greater than that of button and joystick manipulation) "yield a sense of augmented embodiment."[13] So not only are players who use such controllers projecting their sense of self onto a virtual avatar, they are also experiencing an extension of their own selves which allows for even more interactive possibilities.

Regardless of how immersive a game world can be, "the design of a game's interactivity can rely on player expectations and experience, sometimes influencing (or limiting) the design even more than technological constraints."[14] This is particularly true in the case of complex games which require players to learn a multitude of commands and principals before engaging in gameplay. Titles that boast such qualities, such as *Dwarf Fortress* or the *Civilization* series, are

13 Gregersen and Grodal, "Embodiment and Interface," 69.
14 Mark J. P. Wolf, "Assessing Interactivity in Video Game Design," *Mechademia* 1 (2006): 80.

not particularly immersive upon first playing them. Indeed, users who are new to these games are continually ripped from the narrative experience in order to read dialogue boxes filled with meticulous instructions, navigate complex menus, and even search online for help on how to more efficiently play within these titles' limitations. It is only through careful study and continual practice that a player can eventually internalize the complexities found within this genre, a task that must be performed in order for them to reach the aforementioned immersive experience. Even so, it is important to note that this process of learning how to play is part of the interactive experience, and there are many gamers who relish in the task of studying new sets of rules in which they can operate. On the other hand, there are plenty of games which are made up of simple design choices, allowing players to become immersed in the gameplay experience almost immediately. *Pong*, with its straightforward visuals, rules, and goals, offers its players this instantaneous immersion as a result of its lack of in-game distractions and easy learning curve. Having that said, all of these statements are conditional as dependent on a player's previous gaming experiences. A newcomer to the first-person shooter genre will spend more time learning the ebb and flow of such games instead of enjoying the immersive experience they have to offer, while a seasoned user will be able to more quickly adapt to and get lost in those titles which fall under the categories they are most familiar with.

In Michiel Kamp's article on soundtracks and virtual environments, the author argues that "music is a part of a video game's ecology, and thus part of shaping player interaction with the game."[15] As simple as this statement may seem, it expresses a concept which is important for both game developers and soundtrack composers to understand. In essence, music provides players with a context in which to fit their perception of interactivity as influenced by narrative.

15 Michiel Kamp, "Musical Ecologies in Video Games," *Philosophy & Technology* 27 (2014): 248.

Furthermore, an effective score can influence the way a player will choose to interact with a game depending on their prior gaming experiences. Fast-paced themes may cause certain individuals to become more excited or aggressive, while relaxing melodies may soothe others and allow for them to focus on more thoughtful activities such as puzzle solving. In the end, scoring and interactivity go hand in hand as is evidenced by the popularization of music video games in the late 1990s and early 2000s. Indeed, titles which link soundtracks to gaming "foster new opportunities for interacting with music."[16]

## IV. Video Game Performance

These aptly named "music games are … computer games in which music is a main factor of the way the game is experienced or played."[17] This genre (perhaps best represented by iconic titles such as *Dance Dance Revolution* and *Guitar Hero*) brings to mind another important aspect of both game design and the gaming community itself: performativity. Though the topic of video game performance has yet to receive the scholarly attention it deserves, it is a notion which has been present since the advent of interactive entertainment. Whether it was to display the talent required for completing a particular game, demonstrate the proper way to navigate a challenge, or prove one's superiority through competition, performativity has always played an important role in the history of gaming. This is especially true in the case of music games which utilize scoring systems, allowing players to accurately quantify their ability as a "performer". Liebe asserts that the performative aspect of such games inform their parameters and goals, "thus turning the game into a sort of musical instrument."[18] For this reason alone, scholars should be more closely examining the implications inherent to the concept of an individual performing a

16 Liebe, "Interactivity and Music," 50.
17 Liebe, "Interactivity and Music," 41.
18 Liebe, "Interactivity and Music," 42.

video game. This begs the question of which audiences are being reached since this idea affects the way a player perceives interactivity and choice-driven gameplay. In order to find an answer we will turn to research which suggests that the results of these performances have been quite influential on those who seek them, as is indicated by the surge in popularity of online, gaming media. But before we delve into that topic, there are other observations to be made on the subject of video game performance. After all, if we are to discuss performativity then the performable aspects of interactive entertainment need to be made clear.

At a game's most basic level, it is the gameplay itself (that is, the core mechanics of a title's rules and goals) that is most easily performed. Playing through a game as designers intended is the path of least resistance in terms of performable actions, and is also the most obvious way of going about video game performance in general. However, this idea is further complicated by choice-driven games in which players can choose from multiple pathways, ultimately affecting the narrative experience. *The Witcher 2: Assassins of Kings* is a prime example of a title which offers players the freedom to choose in what way its story will unfold. The game boasts sixteen different endings in which certain characters live or die depending on the choices made throughout the narrative experience (see Ex. 1.2, which provides an example of the numerous choices players are presented with during gameplay), giving game participants the option to replay it multiple times as a way of exploring these branching paths. In a case like this, it is the decision-making process that becomes performable. Another way gamers choose to interact with video games is by exploiting or working against a game's rules in order to expose glitches (a process termed “glitching” by the gaming community). A glitch is an unintentional software error which can result in a wide variety of outcomes from a game instantly crashing to a player's avatar attaining immortality. This type of performance highlights an individual's

*Example 1.2. The Witcher 2: Assassins of Kings, CD Projekt, navigating a dialogue menu*




technical skill and knowledge of game exploitation. Players have opportunities to perform for one another at conventions which hold organized gaming tournaments. All types of games are used in these competitive events from fighting games to racing simulators, though certain games have more performative qualities than others. *Dance Dance Revolution* (a music game in which participants are scored on their ability to "dance" in sync with its soundtrack) is often utilized at such competitions, and it has been noted that due "to the possibilities of actually dancing on the mats and improvising movements or performances if the game mechanics are mastered well enough, whole communities with tournaments or show-playing events [have] emerged around the game."[19] Other ways in which a game can be performed are through live shows and "let's play" videos, both of which rely on performers' running commentary (whether it be informative, comedic, or a combination of the two). The term "let's play" (used to describe videos which

19 Liebe, "Interactivity and Music," 43.

document a "playthrough" of a game) has been around since 2007, but the genre was only able to gain traction as a popular form of entertainment in the early 2010s. YouTube celebrities such as PewDiePie (Felix Kjellberg), Markiplier (Mark Fischbach), and CaptainSparklez (Jordan Maron) take full advantage of their ability to entertain audiences with their words and personas while playing popular game titles thus falling under the category of "let's players". However, it is important to note that this form of entertainment is not necessarily limited to digital platforms. As of March 2016, the Game Grumps (Arin Hanson and Leigh Daniel Avidan) have been organizing and performing in gaming performance shows at multiple venues around the United States including the Hammerstein Ballroom in New York City and the Arlene Schnitzer Concert Hall in Portland. And though Simon Gottschalk argued in 1995 that the features of video games are "especially attractive to young people who can now actively participate in electronic spectacles, control them, and master them, rather than passively watch them", no one could have predicted what would emerge from this burgeoning form of entertainment.[20] Giving rise to opportunities for YouTube personalities from the gaming community, the platform's exponential growth would allow for entertainers to stake their claim as financially successful performers. Indeed, much has changed in the past twenty years, and we now live in a world world in which video game performance has grown into a multi-billion dollar business.

## V. YouTube and Gaming

In 2006, it was reported by Rob White that YouTube "is [daily] visited 100 million times and 65,000 clips are uploaded [to its servers]."[21] These extraordinary figures show just how much the platform (designed by a team of former PayPal employees) was able to grow since its

20 Simon Gottschalk, "Videology: Video-Games as Postmodern Sites/Sights of Ideological Reproduction," *Symbolic Interaction* 18 (1995): 5.

21 Rob White, "Treasure Tube," *Film Quarterly* 60 (2006): 3.

launch in 2005, demonstrating the importance placed on both self-expression and entertainment in American society. The website, continuing to grow in popularity, was soon localized in other countries and quickly became one of the internet's most frequented domains. This growth would set the stage for video game performers to emerge as a new force in the entertainment industry. By uploading videos of all genres from the aforementioned let's play to satirical animation, gamers were able to create a niche for themselves in which they could begin laying the foundations for their businesses. In 2013, Gary Weidenaar noted that "two billion videos are viewed daily, with thirty-five hours of video footage uploaded to the site each and every minute."[22] However, as impressive as these statistics were at the time of Weidenaar's publication, the numbers are even higher today.

As of October 2016, "YouTube has over a billion users – almost one-third of all people on the Internet – and every day people watch hundreds of millions of hours on YouTube and generate billions of views."[23] Generating views is synonymous with earning money if a YouTuber is able to both garner a certain amount of attention and regularly produce original content. And due to the nature of the let's play video format, game performers have an incredible advantage in that they have very little work to do in order to produce hours of digital media for their viewers to enjoy. These entertainers simply choose a game (which provides them with an abundance of footage to work with) and then record their reactions and commentary to what they observe on-screen while engaged in gameplay. The majority of let's players choose to play popular games, but there are others who choose to record video series in which they explore more obscure titles. Though the most popular channels tend to stick with the latest and most trending games, it is not necessarily the game choice which dictates a YouTube channel's

22 Gary Weidenaar, "Facebook and YouTube and Choirs – Oh My!," *The Choral Journal* 10 (2013): 58.
23 "Statistics," YouTube, accessed October 14, 2016, https://www.youtube.com/yt/press/statistics.html.

popularity. Instead, it would seem that it is both the personalities of these entertainers and production value of their videos that attract large audiences. This theory accounts for the large margins in viewership between YouTubers (with differences of more than thirty million subscribers between the top two gaming channels) who are actively producing content from the same titles. Regardless, what is clear is that video game performance is now a lucrative business, and recent statistics show that the "number of channels earning six figures per year on YouTube is up 50% y/y."[24] Many of these reported channels are producers of let's play videos, regularly churning out fresh content for their fanbases.

Christine B. Balance has proposed that "in order for a YouTube video to 'go viral,' it must actually incorporate emotional hooks: key signifiers that catch the attention and sensibility of a particular audience."[25] Video game performers fulfill this requirement in that the content they produce is both exciting to new gamers and nostalgic for those who grew up during the rise of interactive entertainment. This allows for such YouTubers to cast a wide net, effectively earning the attention and devotion of viewers from multiple age groups. Furthermore, the casual and improvisatory nature of let's play videos allows for performers to speak freely about whatever topics come to their minds. Although comedy is what makes up the majority of these entertainers' content, YouTube personalities often digress by sharing anecdotes about their own personal struggles before and after becoming a successful let's player. Some performers will even devote time to producing videos in which they thank viewers for their support (usually after reaching a milestone in terms of their channel's subscriber count) or update their fans in regards to major life changes. All of these factors help entertainers gain the viewership of audiences by

24 "Statistics."

25 Christine B. Balance, "How It Feels to Be Viral Me: Affective Labor and Asian American YouTube Performance," *Women's Studies Quarterly* 40 (2012): 139.

using emotional hooks, a process which has proven to be particularly effective in the gaming community. It is also interesting to note how Balance also argued that "these YouTube performances … function as forms of affective labor", that is, work which generates an emotional response.[26] Popular video game performers often speak about their encounters with fans whose lives have been changed because of the uplifting nature of let's play videos, and dedicated fanbases praise entertainers for the positive effects their work produces. However, an inspirational message on its own is not enough to inspire audiences.

Studies imply that "the probability of a YouTube video's 'going viral' hinges precisely on the qualities of authenticity and earnestness."[27] There are many who have attempted to become popular game performers, but few have actually transcended to stardom. And as Balance's research suggests, it may very well be that these lesser performers come across as insincere while recording their let's play and other such gaming videos. The majority of successful YouTubers try to maintain a sense of sincerity while augmenting the parts of their personalities which they feel audiences best connect with. This, as was previously discussed, is what affords let's players the opportunity to connect with their fans on an emotional level, thus reinforcing their viewership. But this digital media platform is not by any means a one way street, and it is important to note that the "key feature of YouTube is not passive viewing but response and dialogue – that is, social interactivity."[28]

Just as gaming includes a number of interactive elements, so does the world of video game performance. In fact, the most popular entertainers are constantly interacting with their audiences through videos, live shows, and social media platforms. Furthermore, fans are engaged

26 Balance, "How It Feels," 140.
27 Balance, "How It Feels," 143.
28 Ayanna Thompson, "Unmooring the Moor: Researching and Teaching on You Tube," *Shakespeare Quarterly* 61 (2010): 338.

in continual dialogue with their favorite YouTube personalities. Other performers find creative ways of connecting with viewers, an example being the Game Grumps playing titles mailed to them by fans. This method has proven to be effective for the comedy duo, and their success has been quantified in the following figures: as of June 2016, the Game Grumps channel had uploaded 4,256 videos, gained 3,418,655 subscribers, and earned a total of 2,414,438,096 views. Due to their regular viewership, the company's estimated yearly income is between $246.4 thousand and $3.9 million.[29] Clearly, this form of entertainment has become something larger than anyone could have predicted, and its interactive qualities are not only important for understanding performance, but also for realizing that gaming itself is centered around interactivity. As supported by both the popularity of let's play videos and digital media, this concept of viewing games through an interactive-centric lens helps scholars to better understand the world of gaming and its music. Moreover, it has been suggested that "games, through music and mutual enjoyment of them from people of all social backgrounds, can bring people together via effective contextualization of such music in a game world."[30]

## VI. On Soundtracks

Jason Brame asserts that "a video game score is the sum of all the music in a game."[31] Indeed, soundtracks have the ability to shape the way a player views the virtual worlds in which they spend their time. Music provides the background for gamers' most exciting adventures, competitions, and moments of creativity. Without a proper score, certain titles would feel empty and lifeless while the genre of music video games would essentially become non-functional. In

29 "Game Grumps," Social Blade, accessed June 27, 2016, http://socialblade.com/youtube/user/gamegrumps.
30 Daniel DeCastro, "Quality Video Game Music Scores, Considering the Standards Set, and Personal Reflections" (master's thes., New York University, 2007).
31 Jason Brame, "Thematic Unity Across a Video Game Series," *ACT - Zeitschrift für Musik & Performance* 2 (2011): 4.

his master's thesis, Pieter Crathorne divided sounds from video games into the following categories: music (the "usually, but not always, non-diegetic" soundtrack), diegetic sound objects ("occurrences which coincide to happenings within the gaming environment"), and ambient sounds (sounds which "appear to be diegetic but whose source can never be found in-game").[32] This type of framework helps scholars to more easily express their thoughts about a game's score and sound objects since it clarifies the main functions of music in gaming. In regards to the non-diegetic portion of a video game soundtrack, we must keep in mind its ability to affect user interaction. Though every player comes to the world of gaming with their own unique experiences and personal preferences, they eventually become accustomed to certain musical tropes found in games of all genres. Suspenseful music is typically used for heightening the senses, indicating that an event is about to take place, energetic tracks often act as a cue for action-packed scenes in which players need to be at the ready, and somber melodies are usually utilized in drawing attention to serious dialogue between a game's characters. Sound objects, which range from the clicking sounds heard when scrolling through menus to the release of an archer's bowstring, provide game developers with a bit of musical "polish" to be used when designing their works. Though these objects are usually mundane in nature, some of them have become iconic such as the "waka-waka" sound made by Pac-Man. The final category is that of ambient sounds, being the most abstract of the three. These noises serve to enhance a game's environment without actually originating from a particular on-screen source, thus falling in between the distinctions of diegetic and non-diegetic sound. Remembering that ambient sounds have no visible source aside from gaming environments helps us to discern them from diegetic sound objects. Examples from this category include the sounds generated by an audible breeze,

32 Pieter Crathorne, "Video game genres and their music" (master's thes., University of Stellenbosch, 2010).

rushing water from an unseen river, or creatures hidden from the player's view. These varying elements can be combined in a variety of ways to create beautiful sound environments which affect the way a player perceives the interactive experience.

It has been argued by artist and theorist Axel Stockburger that "the sounds or objects [from video game soundtracks] are part of a dynamic environment."[33] And since gaming centers around interactivity, it comes as no surprise that sound helps to shape this medium's reciprocal qualities. In-game music itself can change as a direct result of a player's actions, indicating their success or failure at a particular task. In the *Super Mario* franchise, players are rewarded with a short clip of triumphant music for successfully navigating to the end of a level. However, if a player allows for Mario to come to his untimely end (whether it is the result of falling to his death or being devoured by carnivorous plants), they are met with a melody indicative of their defeat. Similarly, adjustments to diegetic sound objects can take place as a consequence of player interaction, often signaling an important event or change in gameplay rules. In 1993 release *Doom* (Ex. 1.3), the protagonist is met with a barrage of unsightly demons who must be destroyed in order to ensure the survival of humanity. When using a less powerful weapon on such foes, players will find that their howls of pain are played at a higher pitch than that of when they are finally destroyed. This change in sound (resulting from multiple instances of interaction) notifies players of an enemy's defeat. Ambient sounds (including silence) may be altered for the purpose of indicating the redundancy of a particular in-game location or a change in narrative. Takashi Tezuka's *The Legend of Zelda: A Link to the Past* begins with the main character making his way through a deluge of rain to the castle just north of his home. Since the sound of this downpour can be heard outdoors (where the raindrops are visible) and inside the castle's ground

33 Axel Stockburger, "The Game Environment from an Auditive Perspective" (paper presented at the Digital Games Research Association, Atlanta, Georgia, 2013).

*Example 1.3. Doom, id Software, obliterating a group of monsters*




level floor, it can be classified as both a sound object and ambient sound depending on the context in which it is heard. After navigating through tunnels connecting the castle to a church sanctuary some distance away, players emerge to find that the rain and its accompanying ambient sound have stopped, signaling the successful completion of the game's first section.

Stockburger has further asserted that a video game's score "consists of a number of sound objects that belong to the nondiegetic part of the game environment", noises which Crathorne would probably choose to classify as ambient sounds.[34] These noises are often found in music video games due to the crossover between their music, gameplay, and use of sound effects. *Guitar Hero*, originally released in 2005 on the PlayStation 2, features gameplay requiring its

34 Stockburger, "The Game Environment."

participants to perform popular songs on the guitar-shaped controlled with which the title is packaged. If players are able to successfully keep in rhythm with the game's various tracks, corresponding pitches will play in conjunction with their well-timed inputs. On the other hand, game participants are met with the dissonant sound of incorrect notes if they cannot manage to press the appropriate buttons within a song's rhythmic structure, further supporting the suggestion that another "purpose of video game audio … is to control player behavior."[35] These sounds of success and failure are not only classifiable as sound objects, but also as part of the title's soundtrack since the two are inextricably linked through gameplay.

## VII. Film Music as Narrative

"Ultimately it is the narrative context, the interrelations between music and the rest of the film's system, that determine the effectiveness of film music."[36] It was Claudia Gorbman who made this claim in her 1980 publication "Narrative Film Music", suggesting that a movie score is only as valuable as its ability to complement the whole of a title's elements. If a film's soundtrack seems out of place, it can throw off the mood or emotional buildup a writer and director had intended for a particular scene. Conversely, a fitting piece of music can add incredible weight to the impact this media can have on an individual. Further expounding on her opening statement, the author proposes that "image, sound effects, dialogue, and music-track are absolutely inseparable during the viewing experience"[37] As it is with video games, visuals play a large role in an audience's perception of a film. Everything from costume design to lighting effects helps to bring a director's vision to life, allowing viewers to take part in whatever world a movie is set. But although there are many similarities between the two art forms, Stockburger has proposed

35 Erik Youngdahl, "Play Us a Song: The Structure and Aesthetics of Music in Video Games" (bachelor's thes., Wesleyan University, 2010).
36 Claudia Gorbman, "Narrative Film Music," *Yale French Studies* 60 (1980): 184.
37 Gorbman, "Narrative Film Music," 190.

that “computer games have a very specific way of deploying sound, which is different from film”.[38] These mediums diverge in that video game music is interactive, a notion which is completely foreign to the art of film scoring.

In gaming, visual aspects function by immersing players in both realistic and fantastical settings. But what is interesting about game imagery is that the ability to portray life-like environments is a relatively new achievement. Indeed, early game artists used simple digital tools for the creation of visuals, resulting in the utilization of symbols to represent objects too complicated to accurately depict. Regardless of which style is considered more popular, it is the responsibility of developers to make sure that the appropriate artistic style (whether realistic or representational) is used to complement a game's overall presentation. Sound effects are another powerful tool employed in films and games of all genres. From the slide whistles of slapstick comedies to the overkill explosions heard in action movies, these diegetic sound objects add to the overall viewing experience. The same can be said for sound effects in gaming since they more or less accomplish the same goals, differing from their cinematic counterparts by the characteristic of interactivity. While films include a fixed set of precisely timed sound objects for the benefit of the narrative's presentation, video game sound effects are nearly always user driven. But this is not to say that gaming sound effects are never used in the same manner as they are in movies. On the contrary, these diegetic sound objects are regularly used during in-game cutscenes (film-like sequences which depict an event) in a set pattern as orchestrated by both game developers and sound designers.

Much is communicated through a film's dialogue such as events, emotions, character relationships, and more. This much is also true in gaming. Just as movies use speech as a means

38 Stockburger, “The Game Environment.”

of exposition, games do the very same thing. However, where gaming once again distinguishes itself from film is in its interactive qualities. Certain titles feature dialogue trees (a mechanism which allows players to choose how they want their avatar to respond during in-game conversations), affecting the overall experience to various degrees depending on a game's design. Released in 2012, *Mass Effect 3* (infamous for its unpopular ending sequence, so much so that an extended cut was released three months later) features a tremendous number of choices in terms of dialogue and actions which ultimately affect the title's gameplay, narrative, and character relationships. This use of dialogue trees not only communicates a story to the player, but also allows players to speak back to the game itself.

Scoring is the final aspect of film which needs to be examined, necessary for its ability to influence all of the previously discussed elements. This sentiment falls in line with Gorbman's ideas, arguing that "the connotative values which music carries, via extratextual cultural determinations and also through textual repetition and variation, in conjunction with the rest of the film's soundtrack and visuals, largely determine atmosphere, shading, expression, [and] *mood*."[39] Music is what brings additional meaning to the works of both film and gaming, capable of shaping the way consumers of these mediums perceive them as a whole. And just as it has been noted that a score can independently possess its own significance, scholar Jessica Green would later expound on this notion by claiming that "audiences can understand the emotions or qualities that music is portraying even when the music is divorced from the image it was created to accompany."[40] This is an interesting remark as it suggests that soundtracks are powerful enough to express meaning without any additional context. Having that said, music needs to be

39 Gorbman, "Narrative Film Music," 203.

40 Jessica Green, "Understanding the Score: Film Music Communicating to and Influencing the Audience," *The Journal of Aesthetic Education* 44 (2010): 82.

analyzed for its ability to complement the viewing experience.

"Through music's development of specific leitmotifs, themes, and cues, the calculated use of film music in conjunction with the other channels of information helps to create the narrative and control the way that the audience interprets a film."[41] Penned by Green in her article "Understanding the Score: Film Music Communicating to and Influencing the Audience", this assertion places soundtracks in a position of power in regards to shaping a viewer's perception. But what is most interesting about the author's suggestion is that the development of music is what works with the rest of a movie's elements. By taking the structural components of a soundtrack and developing them through various transformative techniques, a film composer is able to give even more meaning to a score. As a result, directors are better able to "use music to create and connect the overarching themes of the film and help audiences understand the purpose or meaning."[42] These same techniques can be observed in a number of video game titles, but are most commonly found in role-playing games. In this genre, there is a heavy emphasis on theming, narrative, and character development. *Mother 3*, with its soundtrack by Shogo Sakai, features musical transformation throughout its eight chapters. In particular, the "Love Theme" and "Pigmask Army" themes go through a number of variations throughout the narrative's unfolding, demonstrating the amount of planning which went into the game's design. With that in mind, it is reasonable to argue that listeners "use the score as a tool for understanding the meaning of other channels of information"[43] An effective soundtrack must help viewers to understand a film's narrative, dialogue, and visual aspects if it is to assist them in understanding the overall significance of a script. One of the most powerful tools a composer can use is the

41 Green, "Understanding the Score," 82.
42 Green, "Understanding the Score," 83.
43 Green, "Understanding the Score," 84.

leitmotif, capable of representing characters, locations, and ideas.

These leitmotifs create meanings for listeners by "constructing identifications that are easily recognized within the film", and are most often utilized in dramas.[44] The same technique can be seen in the composition of video game soundtracks, creating the same kinds of associations between music and plot elements. *Final Fantasy VI* (originally released as *Final Fantasy III* in North America) is a title known for its extensive use of leitmotifs and themes to represent each of its main characters and important in-game locations. Because of this, the game's interactive experience is very deep and it's overarching narrative is highly memorable to players who have positive associations with the title's soundtrack. Another way leitmotifs function is by expressing the progression of a film's narrative as they are "modified or altered in order to reflect the changing status of the character, place, situation, or emotion."[45] This too can be observed in gaming as is evident from titles like *The Legend of Zelda: Ocarina of Time* in which players must learn a variety of short themes representative of characters from throughout the game. One such theme, "Zelda's Lullaby", recurs several times over the course of the game's narrative. In each iteration of the short tune, the digital orchestration is slightly altered to reflect changes in circumstances. For example, when Link (the protagonist) first meets Princess Zelda as a child her melody is played on an ocarina as accompanied by a harp and synthesized string

*Example 1.4. Princess Zelda, m. 1-4*



44 Green, "Understanding the Score," 87.
45 Green, "Understanding the Score," 88.

ensemble (Ex. 1.4). Later in the game, when both characters appear to be in their teenage years, the melody of "Zelda's Lullaby" is given to a section of violins as punctuated by harp, strings, and woodwinds. In this version, the figurations in the accompaniment are much more complex thus illustrating the characters' growth and maturity. Green has also implied that another "important role … leitmotif fulfills is to define and distinguish character."[46] The stylistic varieties found in music from around the world can be used to express a cast's personalities, cultural backgrounds, and physical appearances. For better or for worse, composers often associate certain rhythms and harmonies with characters of particular ethnicities as a way of making their identity and temperament clear to audience members. Similarly, video game themes and leitmotifs are utilized to highlight personality traits from a title's main cast of characters. One such title is the 1995 release *Chrono Trigger*, featuring seven playable characters each with their own unique backstories and musical theming. Being the only non-organic cast member, Robo (a humanoid robot) was assigned an uplifting leitmotif indicative of his optimistic personality. Contrary to this is the scythe-wielding Magus (known by the title of lord of the Mystics) whose theme perfectly captures the character's dark and brooding nature.

Ultimately, Green asserts that "music fulfills the more complex role of working … to rhetorically influence the audience's interpretation of the film and the message that the viewer takes from the film."[47] This claim is a recapitulation of the author's previous arguments, summarizing her belief in the suggestive nature of movie scoring. All of the previously discussed elements, whether it be visuals, dialogue, or sound effects, do indeed impact the way in which audiences perceive the intended message of a film. But it is a movie's soundtrack that gives these aspects a context in which to function. As has been pointed out, the parallels between this notion

46 Green, "Understanding the Score," 89.
47 Green, "Understanding the Score," 93.

and the use of music in gaming are numerous. In the next section of my thesis, an exploration on claims made by both students and scholars working in the field of ludomusicology will be presented as a way of preparing readers for my analysis of *Undertale*.

## VIII. Game Music as Narrative

Though the disparities between film music and game soundtracks have already been discussed (the main distinction being the interactive quality of gaming), there is more to be said on the topic of how sound is utilized in video game titles. When it comes to sound effects, their application is generally reserved for one of two purposes. The first of these is to either encourage or discourage player behavior as can be seen in the real-time strategy game *Pikmin*. Throughout the course of this title, players take on the role of Captain Olimar (a human-like alien) who has crash landed on an unidentified planet. Amid his exploration, the captain discovers a race of sentient, plant-based organisms which he names “pikmin”, and then proceeds to cooperate with the creatures for their survival and the repair of his spacecraft. During gameplay, Olimar can treat pikmin by leading them to sources of nectar, providing them with nutrients. The pikmin respond to this with noises of satisfaction and the blossoming of the flower buds growing from their heads. This sonic indication of success (along with visual feedback) helps to reinforce positive behavior helpful to the player's cause. Pikmin can also be sent into combat against the planet's aggressive creatures, and there is almost always a chance of the cooperative lifeforms dying during such an encounter. When a pikmin dies, it lets out a pathetic whimper thus discouraging players (at least, those who sympathize with the friendly creatures) from allowing their companions to get into situations too dangerous for them to handle. Sound effects can also be utilized for the indication of in-game rules (or more specifically, a player's success or failure at a particular task) apart from behavioral conditioning. One such example can be heard in the

1990 release *StarTropics*, an action-adventure title filled to the brim with a variety of puzzles. In the game's fifth chapter, players are made to solve a music-based puzzle before gaining access to the area's main dungeon. This challenge requires the protagonist to play a particular melody (as taught to him by a talking parrot) on a giant pipe organ by stepping on its oversized keyboard. When ineffective in fulfilling this challenge, players will only be met with singular tones produced by the instrument's seven keys. But if the protagonist gives an accurate performance of the aforementioned melody, the organ will play a tune in three-part counterpoint thereby indicating the player's successful completion of the puzzle.

Though diegetic and non-diegetic music are common to both film and gaming, interactive soundtracks are in a distinct category of their own. Consider the *beatmania* series in which players must press a number of buttons and turn a single DJ turntable at the appropriate times in rhythm with the title's various tracks. This music video game responds to a player's inputs to a high degree of precision, actually shifting the position of notes within the game's score according to how accurately a player keeps in rhythm. This participatory aspect (both musical and performative in nature) is absolutely unique to interactive entertainment, and is an example of one of the more interesting applications of music in video games.

Another way sound is employed in gaming is through titles centered around creativity, such as *Mario Paint*. The game, which blurs the lines between gameplay and creation, allows players the freedom to draw, create animations, play mini-games, and work with a music generator capable of producing arrangements of both popular and classical music. In the end, it has been argued that the main "idea behind any successful soundtrack is to adapt the music to match events that occur on screen."[48] Whether it is expressed through the genres of action games,

48 James Lendino, "Scoring for the Modern Computer Game" (paper presented at the International Computer Music Conference, San Francisco, California, 1998).

role-playing games, or freeform games with a focus on creativity, music plays an important role in the shaping of a player's interactive experience.

Axel Berndt and Knut Hartmann have suggested that music is a "narrative medium that does not just express emotions and mood, but becomes a means for the expression of associations and non-verbal comments."[49] This can be heard in film scores, but there are plenty of significant examples of this concept in gaming as well. In 2004 release *Metal Gear Solid 3: Snake Eater*, players assume the role of a fictional CIA agent whose involvement in an international crisis is set during the Cold War. At the start of the title's second mission, it is revealed to the protagonist that he must eliminate his former mentor (a legendary, American female soldier and supposed defector to the Soviet Union). Although the agent struggles with his emotions in regards to completing this task, he eventually completes his objectives and the assassination of the former United States military hero. At the end of the game, the protagonist learns that his mentor was actually loyal to the American government all along, both acting the part of a defector and allowing herself to die at his hands in order to secure international peace. The final scene (in which the agent visits his mentor's grave) is complemented by the track "Debriefing", a work expressive of patriotism, respect, and the protagonist's immense grief. This expression is achieved through the use of recurring motives (associated with these various ideas and emotions) from throughout the game. Berndt and Hartmann have also asserted that a video game soundtrack "connects visually separate locations of a virtual world into a bigger continuous and more believable whole."[50]

The notion of virtual locations being associated with particular themes has also been

49 Axel Berndt and Knut Hartmann, "Strategies for Narrative and Adaptive Game Scoring" (paper presented at Audio Mostly - 2nd Conf. on Interaction with Sound, Ilmenau, Germany, 2007).

50 Berndt and Hartmann, "Strategies for Narrative."

discussed by Crathorne, suggesting that there is often "a connection between a specific gaming environment with specific character(s), and a theme could be used to reference both."[51] To illustrate this point, we will examine 2014 release *Shovel Knight* to find an example of how music can be used as a way of connecting characters with in-game locations. As players traverse the Explodatorium (a dangerous laboratory maintained by the evil Plague Knight) they are greeted with a track titled "Flowers of Antimony (The Explodatorium)". The track in question, making use of chromatic scales, diminished seventh chords, and a whole tone melody, was eventually reworked as a means of devising the battle music which plays during the protagonist's violent confrontation with Plague Knight. Being comprised of motivic transformations of the musical materials from the aforementioned "Flowers of Antimony (The Explodatorium)", this new track was given the name "The Vital Vitriol (Plague Knight Battle)" (Ex. 1.5). When compared with one another, we not only see that the works are in different keys, but that the

*Example 1.5A. Flowers of Antimony (The Explodatorium), m. 33-34*



*Example 1.5B. The Vital Vitriol (Plague Knight Battle), m. 28-29*



51 Crathorne, "Video game genres."

tempo and level of rhythmic activity in the latter track have been dramatically altered. It is this recycling of motivic materials in locations associated with the game's antagonists that generates a sense of consistency throughout the gaming experience. Another role of video game soundtracks is the facilitation of a player's progression through gameplay, and Lendino has proposed that the "balance between textural, ambient music and strong transitional themes is effective in maintaining a consistent feel between the different sections of the game and ensuring that they all connect smoothly."[52] This kind of musical "polishing" can be heard in the critically acclaimed role-playing title *Paper Mario*. The game, released in 2000, is praised for its creative soundtrack. What is relevant to the notion of musical transitions is the way in which the game is divided into chapters. At the presentation of each of these narrative devisions, the first phrase from the track "Mario's Theme" is heard. This subtle use of sound (along with the other ways music is utilized in this title) is what helps this game to maintain its narrative coherence.

Game soundtracks are a multifaceted medium which game developers use for a variety of purposes. They can influence a player's perception and behavior as well as transform the gaming experience into something meaningful. This genre of music plays an important role in the presentation of narrative, suggesting thematic ideas to players not communicated through dialogue or text. Furthermore, soundtracks are clearly capable of being used in conjunction with other aspects of art in order to express anything from emotions to a system of rules. Now that we have looked at a number of techniques which composers use to shape the interactive aspects of video games, I will present my analysis of *Undertale* in light of its utilization of sound and music for the purpose of affecting its players' behavior, perception of characters, understanding of narrative, and awareness of Fox's system of in-game ethics.

52 Lendino, "Scoring."

## *Chapter Two: The Start of a New Journey*

*Undertale* is unique because it gives its players a choice in regards to how they deal with conflict, a choice that ultimately launches them down one of three narrative pathways and vastly different gaming experiences. Indeed, the title allows for a number of decisions which have the potential to alter "the story and tone of the game … in reaction to your choices."[1] As a result, *Undertale* can be performed in a variety of ways depending on the outcomes its players are most interested in. "Good" behavior directs a player toward the "neutral" or "true pacifist" routes while "evil" choices propel them down the "genocide" route. In order to achieve the true pacifist ending, players must have completed a neutral route, befriended every one of the title's main characters, and withheld from killing a single in-game enemy. Contrary to this is the genocide route (achieved by intentionally ending the lives of all creatures and personalities encountered by the protagonist during gameplay), whereas a neutral ending is triggered when a player's actions do not fully satisfy the requirements of either the true pacifist or genocide routes. With that in mind, this chapter will examine the most poignant moments from the first half of the game, comparing those choices with the capacity to significantly alter the gaming experience. I will discuss the context of these choices as well as the way in which *Undertale*'s soundtrack unifies its characters and in-game locations with elements of interactivity.

*Undertale*' s introductory sequence utilizes simple text and images in a dated, pixel art style (Ex. 2.1). The limited palette, utilizing various shades of brown and black, reinforces the idea that the events being described have taken place in the distant past. This technique of suggestion through visuals is comparable to the way in which sepia tones are used to create

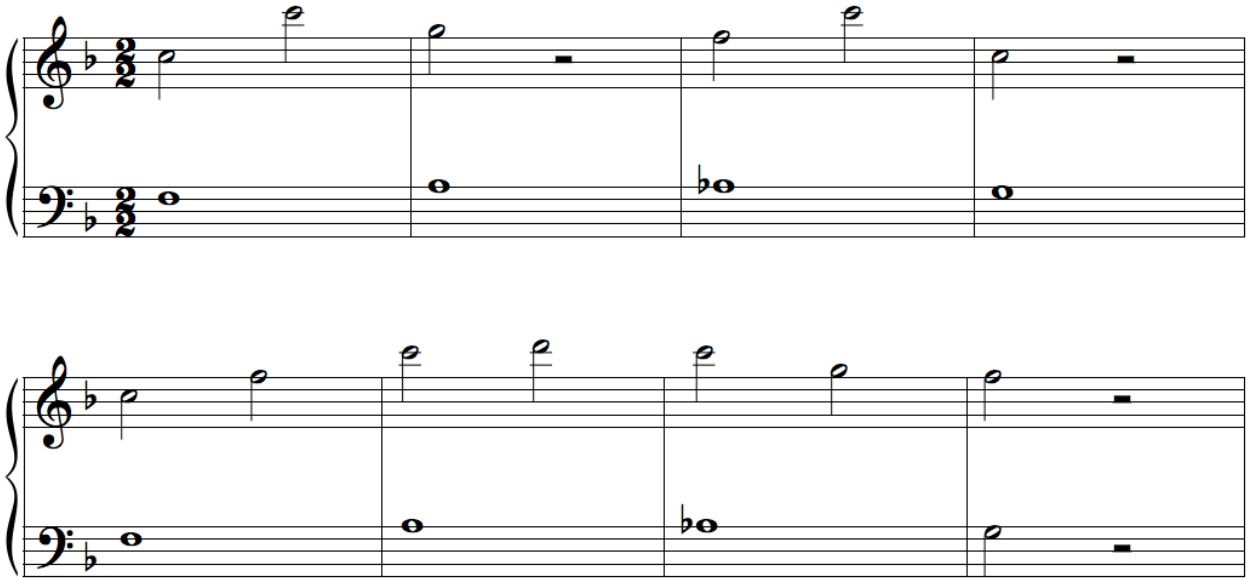
1 "[Fight] or [Mercy]," last modified October 19, 2015, http://geekandsundry.com/fight-or-mercy-challenging-player-morality-in-undertale.

*Example 2.1. Introduction sequence*




associations of antiquity in photography and film. During this opening sequence, the track "Once Upon a Time" (Ex. 2.2) begins to play, complementing the visual presentation with its characteristics drawn from chiptune (or eight-bit) music. The synthesized sounds used in this work are reminiscent to those found in the music of early gaming consoles, and the track makes use of only two and three-part counterpoint during its unfolding of an F major key area with regular inflections to F minor as a result of chromatic passing tones in the bassline. These dated musical characteristics, charming for their nostalgic appeal, further support the idea of this scene having taken place a long time ago. Through the utilization of a slow tempo, players are given a relaxing yet harmonically interesting sonic experience to accompany their intake of the game's background narrative. The track, which is one of the game's handful of non-looping works, then

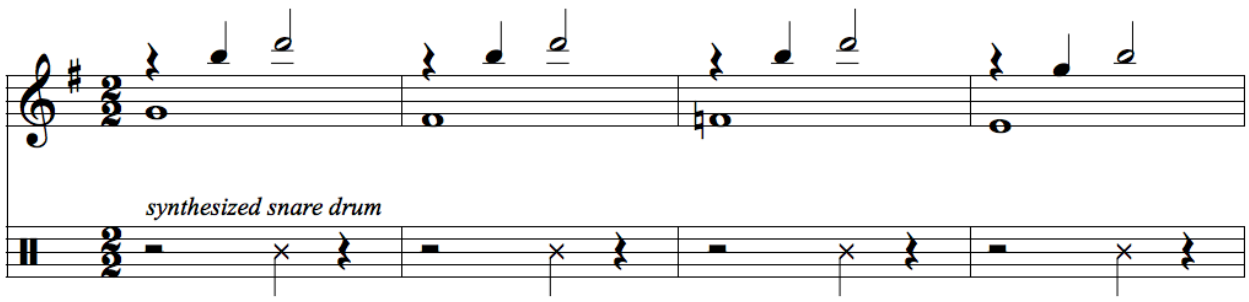
*Example 2.2. Once Upon a Time, m. 1-8*



comes to a close by allowing the simple bassline and synthesized percussion to drop out as the remaining melody decreases in volume until completely silent.

This introductory scene and accompanying music is meant to reference the introductary sequence from Shigesato Itoi's *Mother* (released in 1989) in which the title's narrative is relayed to players through simple text. During this presentation, the track “Introduction” (Ex. 2.3) plays in the background, delivering two and three-part contrapuntal melodies. Nearly identical in style to “Once Upon a Time”, this work utilizes a synthesized snare drum in its opening measures. This obvious nod to Itoi's work is reinforced by the art style, quirky comedy, and narrative

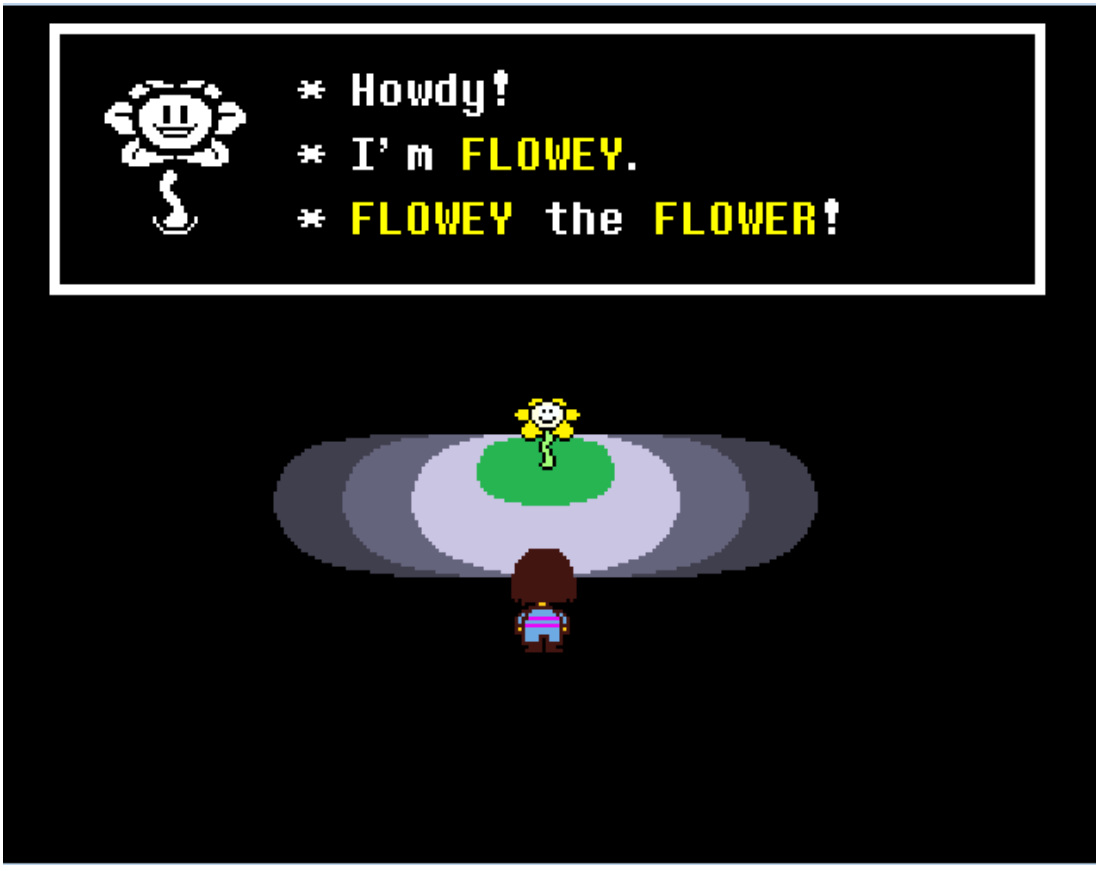
*Example 2.3. Introduction, m. 1-4*



unfolding used in Fox's title. As evidenced by the designer's well-known affinity for the *Mother* series, this referential style helped popularize *Undertale* among fans of Itoi's work.

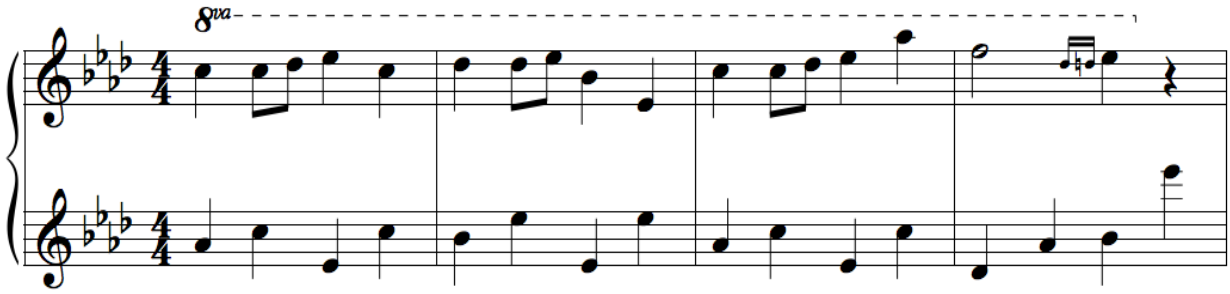
Although video game participants are accustomed to a nearly constant barrage of various noises and exciting music, the absence of sound can be employed for an unsettlingly dramatic effect. By carefully orchestrating these moments of silence, Fox "encourages us to expect the (musical) sound as before, so that when in fact there is no music, we are aware of its absence."[2] This technique is used throughout the game to highlight particular scenes in the overarching narrative, such as the first moments of interactive gameplay in which players are met with no sound whatsoever. At this point, the only available stimuli are the title's pixelated visuals and

*Example 2.4. Encounter with Flowey*




2 Claudia Gorbman, "Narrative Film Music," *Yale French Studies* 60 (1980): 194.

*Example 2.5. Your Best Friend, m. 5-8*




player's control of an in-game avatar (a small, androgynous child) whose singular option is to move on a path leading leftward. Upon following this passage and entering the next area, the protagonist is met with the sight of a single, illuminated golden flower in an otherwise pitch-black room (Ex. 2.4). The sentient flower is found to be smiling, and acknowledges the protagonist with a friendly greeting. The track "Your Best Friend" (Ex. 2.5) then begins to play as Flowey the flower (*Undertale*'s main antagonist) takes a moment to teach the player's avatar a few things about the underground society in which it lives. The work which plays in this scene is a "cute" tune in the key of Ab major, making use of an Alberti bass pattern (as written for what sounds like synthesized laughter) and a simplistic melody (scored for a high-pitched synth lead) comprised of motives heard in later tracks.

This instance of associating a melody with one of its character's is the title's first utilization of leitmotifs, a technique occurring all throughout *Undertale*'s narrative unfolding. In particular, motives from this character's musical theme can be found in a number of other tracks associated with either Flowey or Asriel. Making use of the bassline and melody from "Your Best Friend", the track "Your Best Nightmare" (Ex. 2.6) is what plays as the protagonist confronts Flowey in the final moments of the neutral route. In addition to maintaining the track's speedy tempo, this variation makes use of a harmonization in parallel perfect fifths. We also find this

*Example 2.6. Your Best Nightmare, m. 29-32*




character's leitmotif being utilized in the track "Finale", a work which plays toward the end of the human child's battle with Flowey. What is interesting to note here is that this track is clearly grounded in the key of Ab major (the same key signature from "Your Best Friend") regardless of its regular allusions to F major harmonies. Though there are other examples of this theme being used in subsequent tracks (some of which will be discussed later on), we will continue by examining how the player's initial confrontation with Flowey unfolds.

This poignant scene is used to teach players about *Undertale*'s system of in-game ethics by revealing the nature of the flower's character. To do so, the on-screen visuals make a quick transition to a new window (a gesture which will later be understood by the player as an "enemy encounter") with Flowey at its center, containing a visual representation of the protagonist's soul (a red heart) and an assortment of statistics at its bottom. The creature subsequently explains that the abbreviation "LV" stands for "LOVE", and that by increasing this statistic the protagonist can grow stronger. Then, as a group of five small, spinning ovals rise above his head, Flowey (with a smirk on his face) suggests that LOVE is shared through "Little white… 'friendliness pellets.'" If the player is discerning enough to be skeptical of this offer, they can choose to dodge the projectiles instead of allowing them to collide with their avatar's soul. Otherwise, the protagonist's "HP" (defined as the "level of Endurance that determines the damage a monster or

the protagonist can take before dying"[3]) immediately drops to its minimum value upon contact with the bullets. This first instance of choice informs players of their freedom to perform and explore the results of their in-game decisions. When the player chooses to avoid the projectiles, the flower's countenance immediately changes. Coinciding with this change in attitude is a corresponding shift in key from Ab major to G major in the track "Your Best Friend". This descending modulation betrays Flowey's facade, informing the game participant of his less-than-friendly intentions. Players can choose to evade the bullets twice more (provoking two more descending modulations) at which point the creature becomes completely irate, his face contorting into a frightening teeth-baring grin as the music comes to a complete halt. Projectiles surround the protagonist's soul in an unavoidable, slowly constricting circular pattern as the bloodthirsty monster bursts into a fit of maniacal laughter. But just as the player's avatar is about to be destroyed, they will be rescued by an anthropomorphic, fire-wielding goat.

Thanks to the actions of Toriel (otherwise known as Goat Mom by the *Undertale* community), Flowey is momentarily dispatched. Then (in taking the space the flower once occupied) this new character moves to the center of the screen while commenting, "What a terrible creature, torturing such a poor, innocent youth…" It is in this moment that the juxtaposition between good and evil is made clear to players as the track "Fallen Down" (Ex.

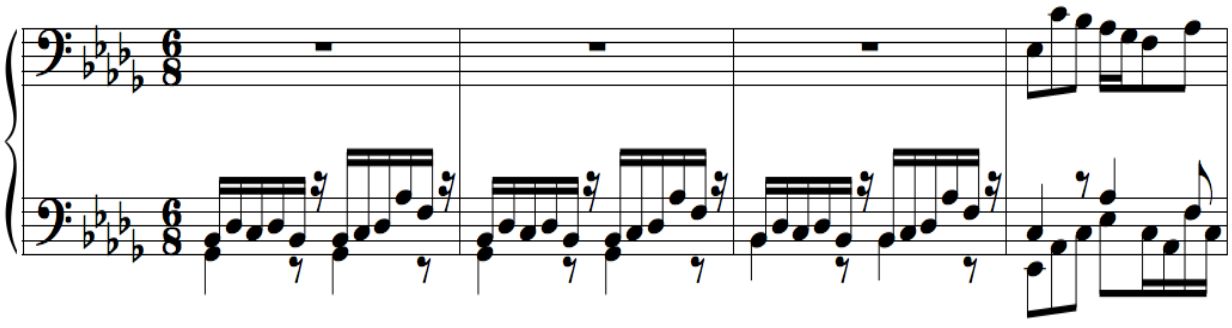
*Example 2.7. Fallen Down, m. 1-4*



3 "Stats," last modified May 16, 2016, http://undertale.wikia.com/wiki/Stats.

2.7) begins playing in the background. The ethical implications here are obvious, as a clear cut binary between good and evil is presented through the juxtaposition of Flowey's actions and those of the character who just saved the protagonist. Even the musical contrast between the two characters is telling. While the flower's theme, given the ironic name “Your Best Friend”, expresses a sense of insincerity with its descending semitonal modulations, “Fallen Down” is a gentle piece in triple meter scored for piano as accompanied by warm synthesized sounds. The track's major seventh and non-functional dominant ninth chords give it a dreamy quality, as if it were a lullaby. While this particular theme is only utilized in a small number of tracks (the goat's leitmotif appears in “Fallen Down (Reprise)” and “Bring It in, Guys!”), Toriel is one of the few characters in Undertale who have more than one melody associated with them. When in combat with this character, players hear the track “Heartbreak” (Ex. 2.8). With its bassline and lower voice counterpoint in compound meter, this work uses a variety in its orchestration ranging from eight-bit sounds to modern orchestral instruments. Furthermore, the number of voices being utilized and orchestral texture change often enough to keep the work interesting from beginning to end. Like many other pieces from Undertale's soundtrack, “Heartache” makes use of a B section which introduces new melodic material for use in later tracks. This work's motives are later reused in the music played during the protagonist's confrontation with Asgore, linking the

*Example 2.8. Heartache, m. 1-4*



*Example 2.9. Ruins, melodic transcription, m. 5-17*




two characters together through motivic transformation.

As players follow their newly found savior north of the room in which Flowey was lurking, they enter the Ruins. It is here that a track named “Ruins” (Ex. 2.9) starts to play, beginning with solo piano. This instrumentation harkens to the scoring of Toriel's gentle theme, immediately giving the work a positive connotation. “Ruins” (making use of the dorian mode) is non-threatening in its harmonies and rhythms, instead providing players with an encouraging sonic atmosphere during the portion of gameplay in which puzzle solving and conflict resolution emerge as core elements of *Undertale*'s interactivity. During this scene, Toriel patiently guides the protagonist through a variety of puzzles while offering words of encouragement such as “I am proud of you, little one.” But because the player's in-game avatar is a human child, the motherly goat warns them about the likelihood of being attacked by monsters. The guide then encourages players to “strike up a friendly conversation” if they are confronted by a hostile monster as a means of stalling for time until she can come to their rescue.

The protagonist is then given an opportunity to practice this newly acquired skill on an

*Example 2.10. The dummy scene*



*Example 2.11. Anticipation, bassline, m. 1-4*

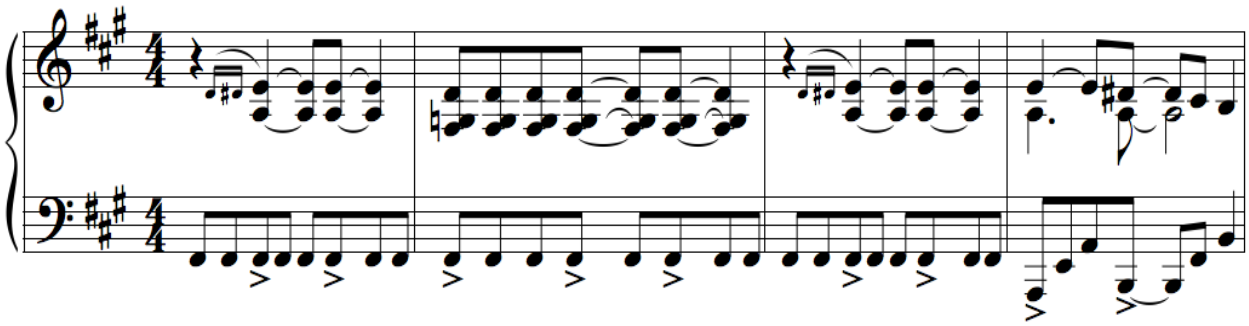


immobile dummy (Ex. 2.10), a scenario in which players are tested on their willingness to obey Toriel's requests. Ultimately, this scene is used to further reinforce *Undertale*'s thematic use of morality, and just as we observed in the confrontation with Flowey, a quick visual transition is made to the enemy encounter window with as the track “Anticipation” (Ex. 2.11) begins playing in the background. Considering the low-stakes scenario of chatting with a practice dummy, the accompanying music is rhythmically frenetic. This contrast (a result of Fox's sense of comedic

irony) not only serves to reinforce the disparity between violence and pacifism, but also heightens the sense of tension in regards to what decision players will make in this situation. Unlike in the previous encounter with Flowey, the protagonist is now given a choice between fighting, acting (that is, observing, conversing, and other such options), using an item, and showing mercy (either by running away or sparing the helpless dummy). This instance of performative decision-making allows a wider range of consequences to be explored in this "combat" situation. If the player is obedient to their guide's request and begins talking to the dummy (who "doesn't seem much for conversation"), they are rewarded with the words "Ah, very good! You are very good." Once again, non-violent behavior is strongly encouraged in this early part of the game in order to make a distinction between good and evil. Furthermore, this disparity is directly associated with a player's willingness to submit to in-game characters as a means of receiving praise. However, if the protagonist engages the dummy with violence, they find that it is easily destroyed. Toriel then chides the player and kindly asks, "We do not want to hurt anybody, do we…?" It is important to note that a player's decision here does not affect which route (whether true pacifist or genocide) the narrative begins to aligns with. Instead, this situation is used to teach players what the game expects of them. While peaceful behavior is commendable, violent actions are worthy of being reprimanded.

In only a few minutes do players reach a point in which the protagonist is left to wait for Toriel's return, finally being given the freedom to explore the Ruins at their own pace. One of the first things a player can do with their newfound liberty is talk with the frog-like creature in the next area. This small monster offers the protagonist further advice on how to avoid violence by pleading with them to "Use some MERCY, human." It is clear from this appeal that the title's system of in-game ethics is only further encouraging non-violent behavior, a theme which

*Example 2.12. Enemy Approaching, m. 1-4*



permeates the entirety of the pacifist route. While navigating from room to room, players will begin experiencing random enemy encounters (once every twenty to thirty seconds) as they solve puzzles and search for an exit. However, since Toriel is no longer around to provide protection, the protagonist is left to diffuse these violent situations on their own. It is during such encounters that we hear the track "Enemy Approaching" (Ex. 2.12). This track is the complete realization of "Anticipation", making use of its predecessor's bassline and rhythmic percussion. It is the melody, upper voice counterpoint, and synthesized orchestration that give its musical texture an exciting quality, complementing the intensity of these battle scenarios. Additionally, the track utilizes a B section in order to extend its use in gameplay.

During such an encounter, the protagonist is given the same choices they had when practicing with the dummy: they can fight, act, use an item, or show mercy to their opponent. As a means of providing added support for *Undertale*'s inclination toward pacifism, the "ACT" option is made more enticing by presenting the player with a variety of options for them to choose from depending on the monster with which they are engaged. Indeed, "by saying or doing the right things, you can sap away at an opponent's will to fight, then choose Spare to resolve the matter peacefully."[4] And as long players continue to deal with confrontations in this manner,

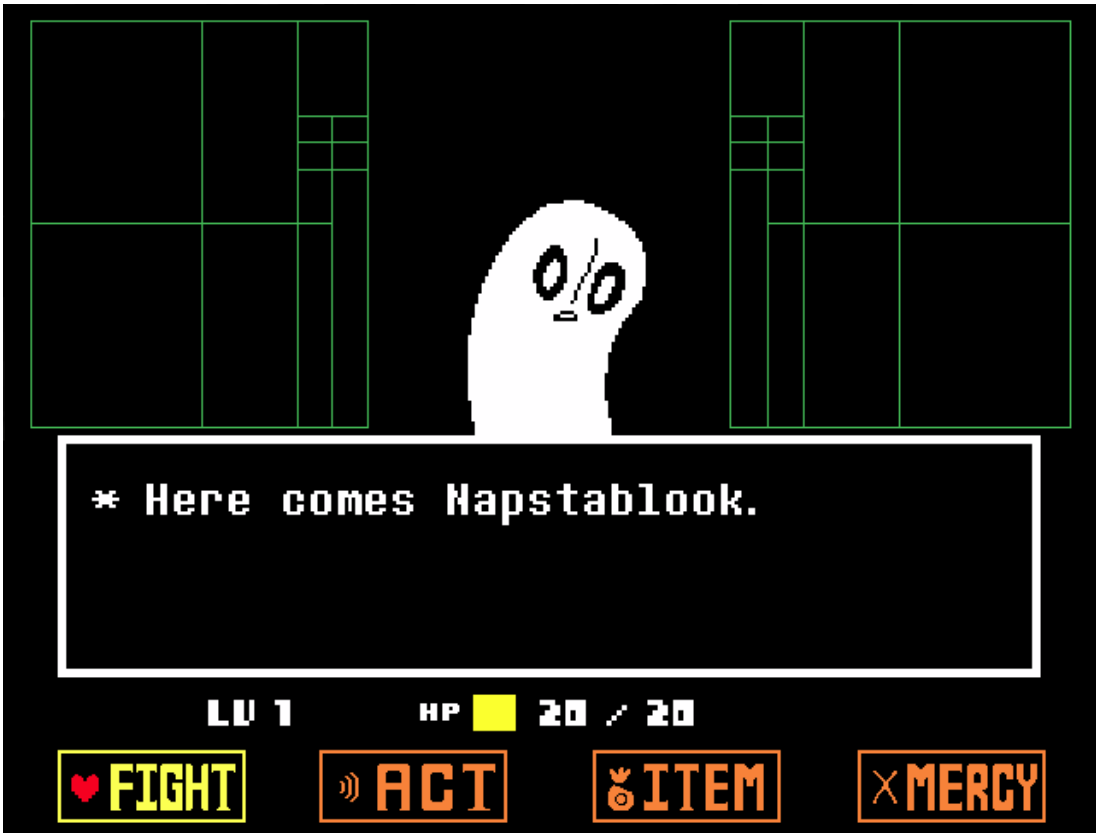
4 "Undertale's Not as Peaceful as it Pretends to Be," Kill Screen, accessed May 4, 2016, https://killscreen.com/articles/undertales-not-peaceful-it-pretends.

they can remain on the pacifist route. In contrast to this harmonious ideal is the "FIGHT" command, which allows the protagonist to physically harm their opponents. Players are not rewarded with visual variations as a result of choosing to kill their enemies, but are instead given gold and "XP" (a common abbreviation for "experience points" in most role-playing games, but later revealed to signify "execution points"). Indeed, each of these encounters can be simplified to the singular decision between pacifism and violence, and every "battle, from a random encounter to a major boss, is an opportunity to show mercy or to kill".[5] Moreover, it is often the case that a choice between two contrasting actions (such as "Complement" and "Threat" or "Console" and "Terrorize") is given to the player, further juxtaposing the notions of good and evil. Other pairings, such as "Imitate" and "Flirt", serve as comic relief due to their impractical nature. Regardless of the decision they make, players are given a wide variety of choices whose results can be explored through experimentation, and actions which yield an interesting or comical result can be repeatedly performed as long as the conflict between the protagonist and monsters in question does not come to an end.

Amid the protagonist's exploration of the Ruins, they will eventually find and be forced to confront Napstablook the depressed ghost (Ex. 2.13). Here, Fox uses an emotionally sensitive character to give players an opportunity to either practice empathy or violence. This contrast, made clear in the way the battle unfolds in response to a player's decisions, begins to shed light on the differences between the true pacifist and genocide routes. The monster in question (a white ghost with similar stylistic traits to the ghosts from *Pac-Man*) is obstructing the protagonist's path, though not necessarily with any malicious intent. In fact, it is revealed to players that the ghost is only pretending to sleep in order to avoid social interaction. During the

5 "[Fight] or [Mercy]."

*Example 2.13. Fight with Napstablook*




*Example 2.14. Ghost Fight, m. 1-4*

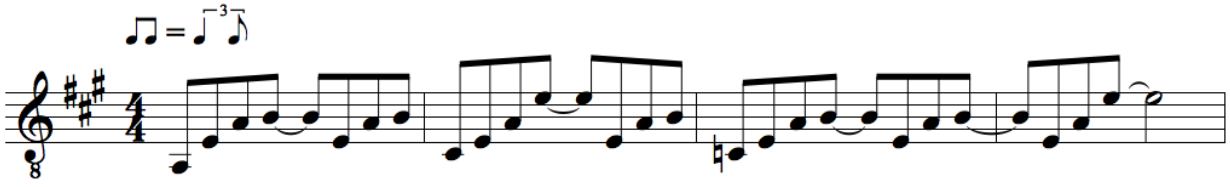


encounter, the track "Ghost Fight" (Ex. 2.14) begins to play as players attempt to resolve the conflict. This music creates a silly atmosphere with its strange orchestration, walking bassline, and bombastic soli sections. In the pacifist route, the conflict with Napstablook is resolved after the player tries to cheer up the ghost four times. Each time the spirit is encouraged, its "attacks"

(tears falling from its eyes which act as projectiles) decrease in intensity, and the ghost finally says to the player, “i usually come to the RUINS because there's nobody around… but today i met somebody nice… ” When completing the genocide route, Napstablook will fade away and disappear upon sighting the protagonist if the player has already murdered each of the ruin's inhabitants. Otherwise, the spirit can be engaged in combat and “defeated” with the FIGHT command. At the end of the battle, Napstablook admits that ghosts cannot be killed and that they were only lowering their HP out of courtesy, then suggesting to the protagonist, “pretend you beat me… ooooooooo”. Afterward, the player is rewarded with neither gold or XP, but is instead informed of their loss of one experience point. This subtle clue allows players to see that there is a difference between experience points and XP, as the protagonist's number of execution points goes unaffected by the aforementioned reduction.

When the protagonist arrives in front of Toriel's home, the motherly goat is surprised to see them and apologizes for her absence. The two then enter the modest house as the track “Home” (Ex. 2.15) begins playing gently in the background. Comprised of musical materials taken from the game's opening work (“Once Upon a Time”), “Home” is scored for solo acoustic guitar. This reuse of motivic structures can be observed by comparing the upper voices from each of the two tracks (Ex. 2.16). What sets this work apart from its predecessor is the gently ebbing and flowing accompaniment. The lower voice counterpoint, composed in an idiomatic style for acoustic guitar, creates an atmosphere of peace and relaxation appropriate for Toriel's

*Example 2.15. Home, m. 1-4*



*Example 2.16A. Once Upon a Time, m. 1-4*



*Example 2.16B. Home, m. 17-20*



dwelling place. During this scene, the player is told that this will be their new home and that they are to have “a nice time living here.” The protagonist is even given their own room by the motherly figure, whose intentions seem to be deeply rooted in caring for others. Should the protagonist take a nap in their new bed, the musical background will switch from “Home” to “Home (Music Box)” (which utilizes a synthesized music box instead of acoustic guitar). Since the two tracks are comprised of identical materials, they can be interchanged at any moment without a loss of either rhythmic or melodic continuity.

These circumstances, along with their musical accompaniment, are meant to endear the maternal goat to *Undertale*'s participants. Indeed, this tactic of using pleasant musical themes in association with the title's characters is seen all throughout the gameplay experience. When spoken to, Toriel tells the protagonist how glad she is to “have someone here” and that she has “prepared a curriculum” for their education. Players are then allowed the choice of asking the question “When can I go home?” to which Toriel replies, “This… this IS your home now.” As

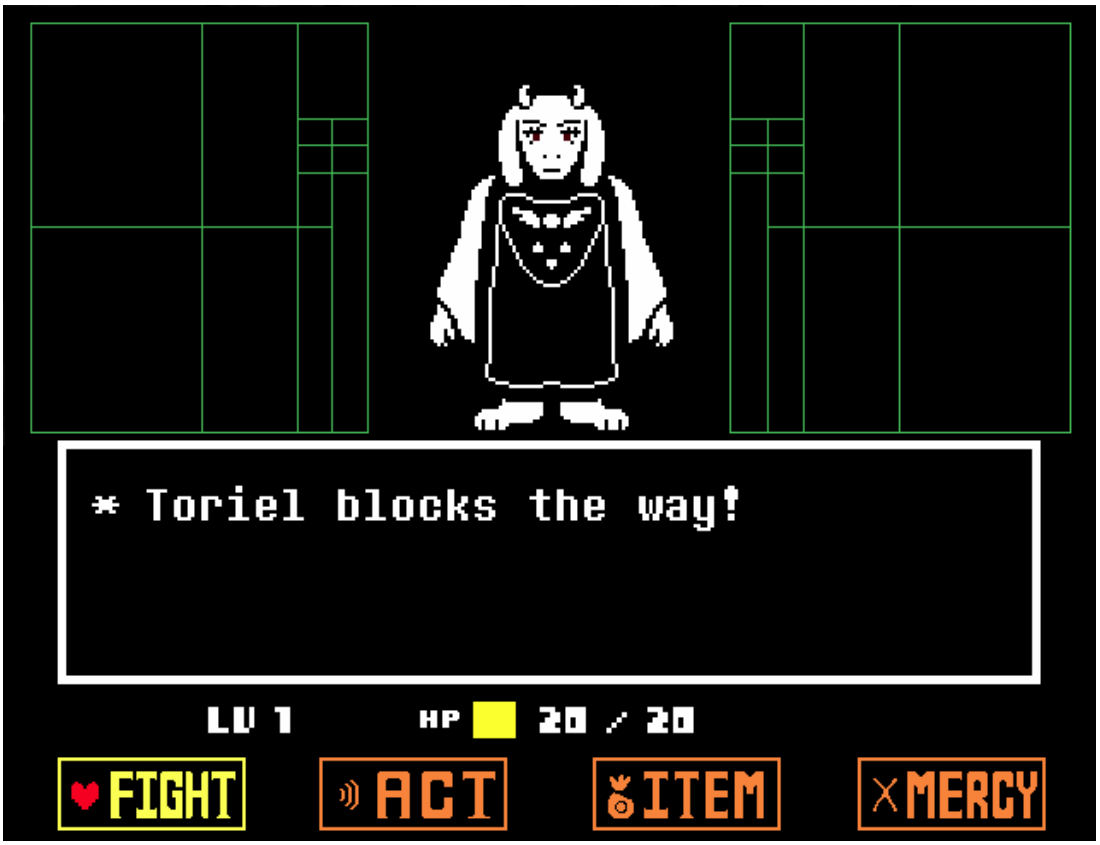
nice as everything seems in this picturesque scene of home cooked meals and motherly love, the protagonist is still lost in an unfamiliar world, separated from whatever life they had on the surface. Moreover, this dismissive response from Toriel is somewhat disturbing. It reeks of manipulation and an intent to take away the protagonist's freedom, thus putting the goat's character into question. If the protagonist happens to repeat the question twice more, Toriel will leave the room while suggesting that she has to "do something." It should be noted that choosing to persist in this line of questioning is not really a choice since doing so is the only means of progressing the game's narrative. Afterward, players can explore the premises until they reach the basement, at which point the background music ceases and is replaced by an atmospheric, intermittent, and low-pitched percussive sound. This stark contrast in sonic presentation creates a mood which complements the brief monologue delivered by Toriel:

> You wish to know how to return "home," do you not? Ahead of us lies the end of the RUINS. A one-way exit to the rest of the underground. I am going to destroy it. No one will ever be able to leave again. Now be a good child and go upstairs.

Once again, the player is given a choice. Do they obey the command they were given, or do they follow through with their intentions to return home? It is situations like these which further confuse the issues of morality and obedience portrayed in *Undertale*. In its defense, the title is hinting to players that something is awry by the deliberate change in background music. This shift in atmosphere, in addition to Toriel's seemingly uncharacteristic behavior, is meant to help the player understand that this is a crucial decision-making moment. But much like before, this presentation of choice is somewhat misleading since the only way to move forward is to pursue a way out of the Ruins.

Upon following Toriel further into the basement, it is revealed to the player that they are

*Example 2.17. Confrontation with Toriel*




being kept from the rest of the Underground for their own safety. Indeed, the now stoic goat informs the protagonist that she has seen a number of lost children leave her home only to die in the dangerous environment ahead. If the player insists by having their avatar navigate to the end of this area, Toriel will confront the protagonist (Ex. 2.17) while demanding, “Prove to me you are strong enough to survive.” Everything has been leading up to this moment. The juxtaposition between good and evil, Toriel's characterization, and the way in which players have been rewarded for their non-violent behavior are being tested in this pivotal confrontation which ultimately determines whether or not the protagonist will shed blood. This performative confrontation begins with the enemy encounter window coming into view as the track “Heartache” begins playing in the background.

Whether it was intended by Fox or not, the complexity of this track complements the puzzling nature of the confrontation with Toriel. Indeed, players who want to avoid violence will find that talking is ineffective and that fleeing only further convinces the goat that the protagonist is incapable of surviving in the Underground. The only non-violent means of moving beyond this point in the game is to dodge an onslaught of magic fireballs while repeatedly using the "Spare" command (at first appearing to have no effect) until the motherly goat becomes emotionally overwhelmed by the protagonist's peaceful approach to combat. This in turn motivates Toriel to allow a player's progression into the next in-game area despite her fears. Ultimately, this peaceful solution to combat is not particularly intuitive, and the "constraints of *Undertale*'s nonviolent combat, then, might suggest something similar, highlighting the way that choosing not to fight is a profound risk, not to be taken lightly."[6] Once the confrontation is at an end, players are rewarded for their pacifism with the ability to progress, a hug from Toriel, and a reprise of "Fallen Down" as a means of encouraging their behavior. And so, the stage is set for the rest of the game. Although the world of *Undertale* is full of perils, our protagonist can avoid resorting to violence if they truly possess determination.

This notion of choosing between violence and pacifism is then tested by the title's gameplay elements. Indeed, players begin to experience the consequences of their actions (in regards to changes in the title's narrative and soundtrack) immediately after the protagonist's confrontation with Toriel. In the next area, that is, Snowdin Forest (Ex. 2.18), players are introduced to a couple of *Undertale*'s most iconic characters (Sans and Papyrus, two of the three skeleton brothers), and are then free to explore the environment while the track "Snowy" (Ex. 2.19) plays softly in the background. Making use of piano, strings, and piccolo, this delightful

6 "Undertale's Not as Peaceful as it Pretends to Be."

*Example 2.18. Snowdin Forest*



*Example 2.19. Snowy, m. 9-12*



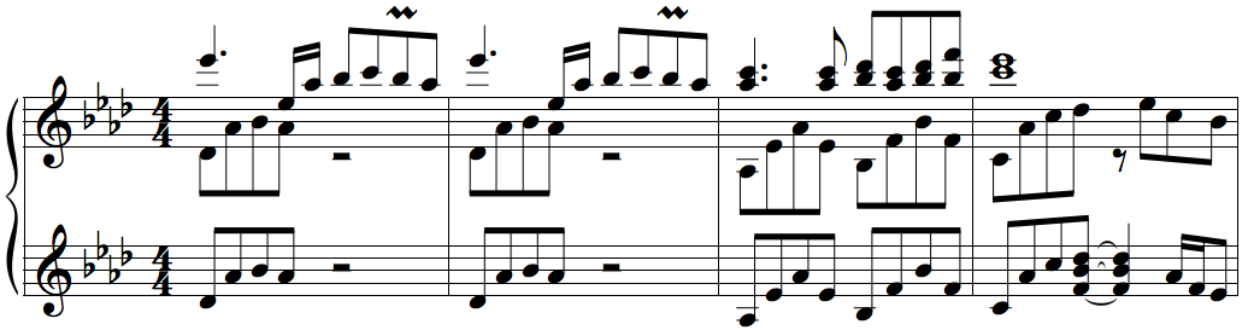
tune complements the winter landscape which the protagonist must traverse in order to to reach their next destination. The work's high registration reflects the chilly atmosphere of Snowdin Forest while the staccato articulations in the piano part evoke imagery of gently falling snowflakes. Furthermore, its motivic materials are later reused in a number of other tracks,

giving this theme significance in the title's overarching narrative. For a player who has thus far met the requirements for the genocide route, they will find that the track “Snowy” plays at a noticeably slower playback rate (though not to the point of distortion). Since this indication is only noticeable to those players who have already navigated Snowdin Forest while on the neutral or true pacifist routes, it serves as a warning for them to cease in their violent tenancies. This is even reflected in dialogue as seen in one of Sans' most quoted lines: “if you keep going the way you are now… … you're gonna have a bad time.”

Much like in the ruins, players are continually met with random enemy encounters as they travel on foot, once again being given the choice of reacting with either violence or pacifism. Though players can never go back and undo the actions they have taken in the Ruins (at least, not without restarting the game from its beginning), it is important to note that their performance from this point onward can affect the outcome of their playthrough. Indeed, if a player on the genocide route fails to murder each and every inhabitant of Snowdin Forest, they will be permanently shifted back to the neutral pathway on which they first started. Similarly, for players who aspire to witness *Undertale*'s true pacifist ending, a single accidental killing of one of the forest dwelling monsters will place the protagonist onto the neutral route. As for those already on the path to the neutral ending, there are no actions they can take which will move them onto the true pacifist or genocide routes at this point in the gameplay experience.

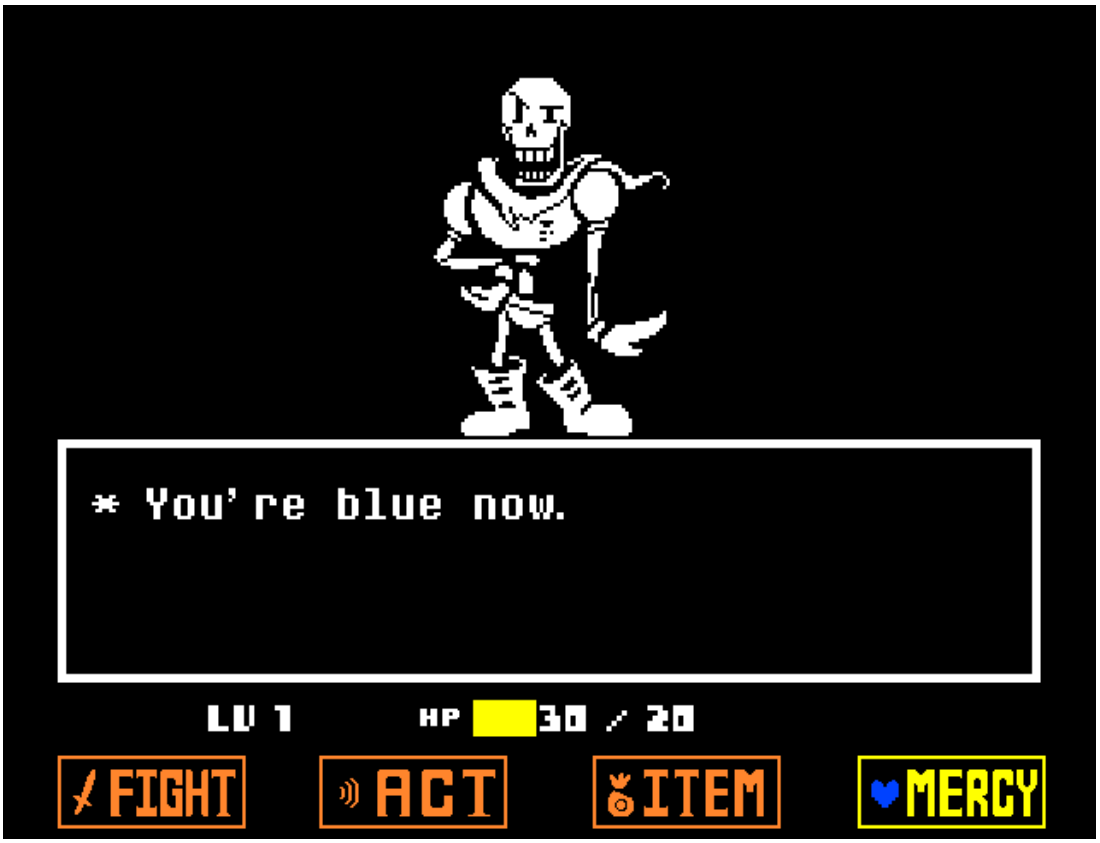
Fox uses music to not only portray the title's characters in a certain light, but to reward players for staying within the boundaries of *Undertale*'s system of in-game ethics as well. This can be observed upon the player's arrival in Snowdin (a small, winter themed town) in which the track “Snowdin Town” (Ex. 2.20) begins to play as the protagonist wanders throughout the village in search of items, lodging, and information. The track, making use of piano, strings, and

*Example 2.20. Snowdin Town, m. 9-12*

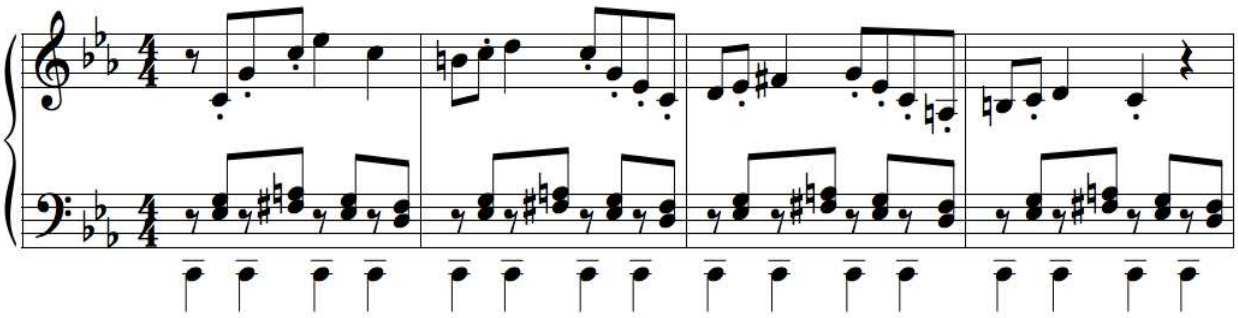


some kind of metallophone instrument, serves as a warm welcome to those players who have refused to exterminate the citizens of the Ruins and Snowdin Forest. It utilizes the same rhythmic and motivic materials as "Snowy" but at a slightly faster tempo, giving this work a lively and innocent quality. Upon leaving Snowdin to search for their next destination, players are met with

*Example 2.21. Battle with Papyrus*



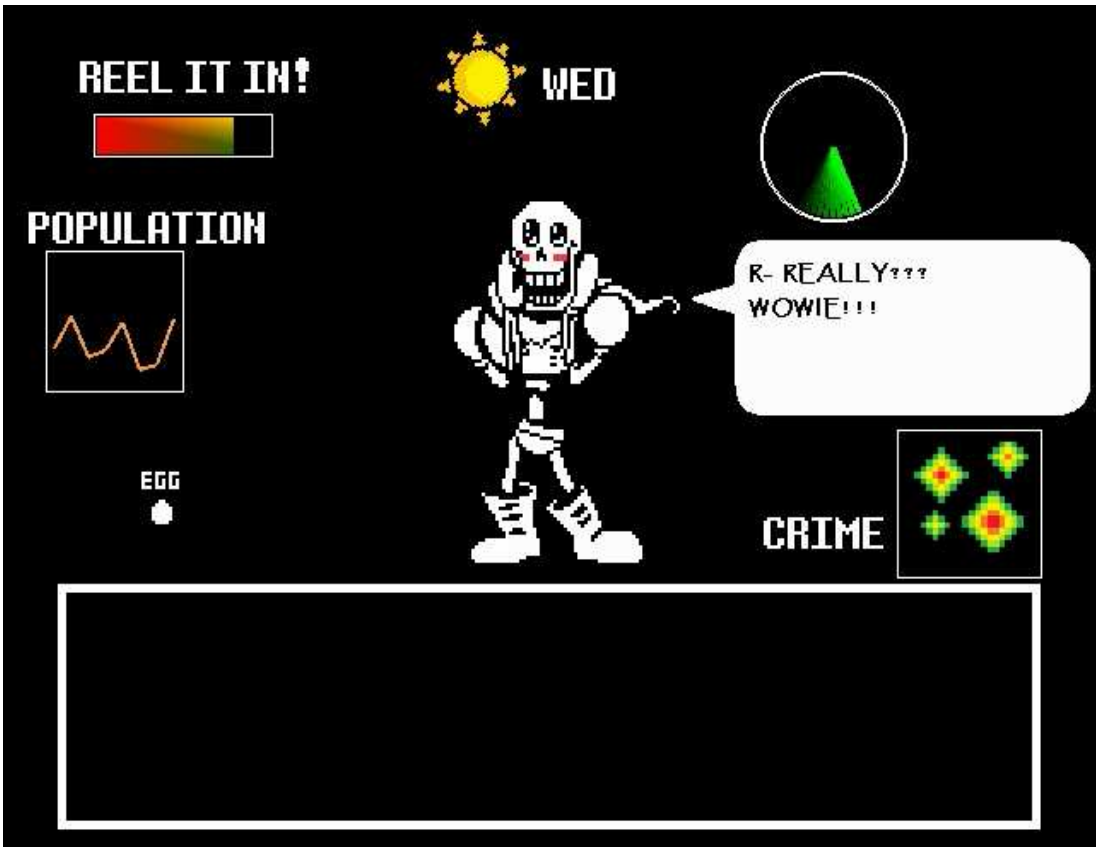
*Example 2.22. Nyeh Heh Heh!, m. 5-8*



yet another interesting moment in gameplay: the protagonist's "battle" with Papyrus (Ex. 2.21), an encounter which can either be humorous or macabre depending on which narrative path a player is currently on. In the neutral and true pacifist routes, players are confronted by the skeleton as the enemy encounter window opens with the track "Nyeh Heh Heh!" (Ex. 2.22) playing in the background. This silly work is a sonic representation of Papyrus' nature, complementing the character's strange personality and supposed lack of malicious intent toward the human child. Orchestrated with eight-bit synthesizers, this track moves at a quick tempo as the protagonist deals with the first portion of this combat scenario. In the second half of this battle, the track "BoneTrousle" begins to play in the background as players are forced to dodge a variety of moving, bone-shaped obstacles. This work is an amplified version of "Nyeh Heh Heh!", making use of additional counterpoint, an intensified percussion part, and complex synthesizers (as opposed to only utilizing eight-bit sounds). This variety in sonic presentation is what players are rewarded with as long as they avoid or fail to meet the requirements of *Undertale*'s genocide route.

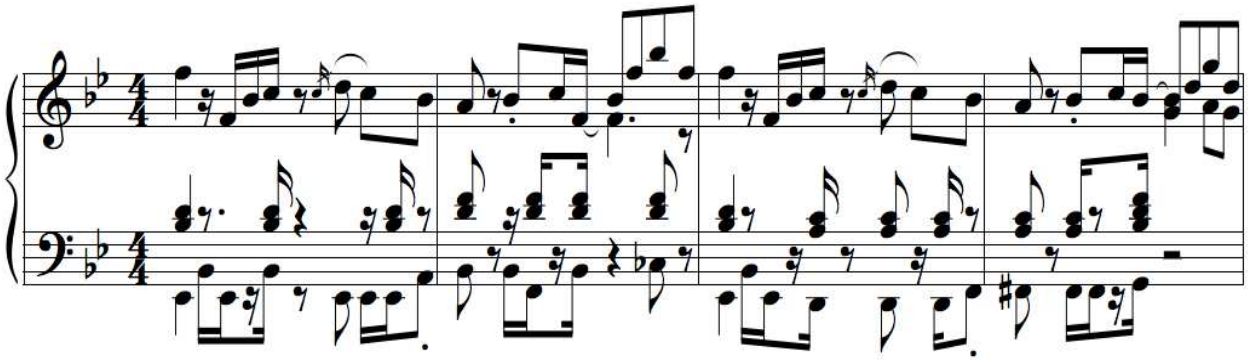
Yet another example of how players are rewarded for choosing pacifism is the notion of additional content. In short, there is much more to see and experience on the true pacifist route as compared to the genocide route. For those who chose to spare Papyrus, they are given the

*Example 2.23. Date with Papyrus*




opportunity to visit him at his home in Snowdin, and if the protagonist has flirted with the skeleton during their battle he will take them on a date (otherwise, the characters engage in a "hangout" session). During this scene, the enemy encounter window makes an unexpected appearance as the words "DATING START!" appear in the center of its text box (Ex. 2.23). What follows is a comical sequence of antics intended to parody various design elements from dating simulation games, thus serving as a humorous gag for fans (and critics) of the genre. This ridiculous shift to a pseudo-romantic scenario is accompanied by the track "Dating Start!" (Ex. 2.24) which utilizes several motivic materials from "Snowdin Town". Making use of both eight-bit sounds and other synthesizers, this work evokes an air of innocence and joy with its moderate tempo and colorful instrumentation. Furthermore, its lively rhythms make for an interesting

*Example 2.24. Dating Start, m. 1-4*

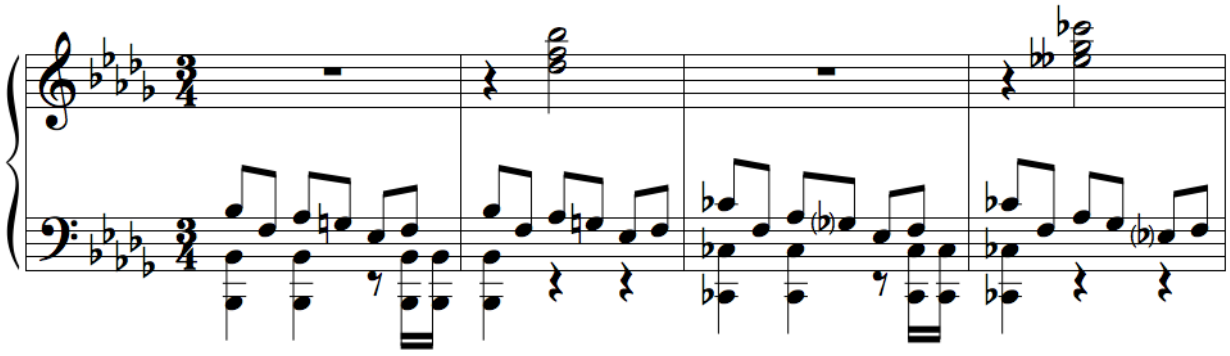


variation on the motives used for the track's composition. As the "date" progresses Papyrus becomes more and more uncomfortable with his inexperience as a romantic, escalating the overall dramatic tension as indicated by a switch in soundtrack to "Dating Tense!" as followed by "Dating Fight!". While the first of these two tracks makes use of a new melodic profile, the latter is comprised of musical ideas from both "Snowdin Town" and "Dating Start!". Ultimately, this scene (only available to players on the neutral and pacifist routes) is part of what gives *Undertale* its unique charm, inspiring Kallie Plagge of IGN to record the following observation in her review of the title: "When I was trying to go for a more violent run, fighting monsters I'd once flirted with made Undertale's message about humanity hit even harder."[7]

Once finished with Papyrus, the player is introduced to the second of *Undertale*'s serious antagonists. Up to this point in the game, the protagonist has come in contact with an array of monsters who can be pacified by means of non-violent actions. But the following circumstances signify a departure from the title's whimsical aesthetics as players learns about Undyne, an armor-clad soldier who has only one purpose in mind: to find and kill a human in order to take their soul. In this scene, the player narrowly escapes the warrior's searching eye by hiding in a field of tall grass as the track "Undyne" (Ex. 2.25) plays ominously in the background. The work

7 "Undertale Review," last modified January 12, 2016, http://www.ign.com/articles/2016/01/13/undertale-review.

*Example 2.25. Undyne, m. 1-4*



(consisting of motives utilized in other tracks associated with the antagonist) is both haunting and threatening, coinciding with Undyne's ferocious character. Its march-like tempo and use of piano, strings, timpani, and percussion give it its sinister quality as players begin forming their opinions of the mysterious soldier.

The way Fox uses these musical elements to tie atmosphere and personality together is similar to the consistent theming found in the Final Fantasy series, in which composer Nobuo Uematsu uses motivic transformation as a means of connecting characters and in-game locations.[8] An example of this can be observed during the protagonist's exploration of the Waterfall region, an in-game location accompanied by a track designed to complement the area's gloomy atmosphere. Being comprised of musical materials later identified with Undyne, "Waterfall" (Ex. 2.26) helps to create a relationship between the antagonist and current gaming environment. In particular, what gives this track its dismal quality is its sluggish tempo, low registration of the piano, and counterpoint of the glockenspiel's part.

It is in the following scene where a genocide playthrough strongly diverges from the neutral and true pacifist routes, marking a pivotal moment in gameplay and narrative. After the player's avatar has worked their way through the Waterfall region, they are confronted by a

8 Pieter Crathorne, "Video game genres and their music" (master's thes., University of Stellenbosch, 2010).

*Example 2.26. Waterfall, melodic transcription, m. 9-16*



young monster. This character (known as Monster Kid) then speaks to the protagonist, approaching them with a different monologue depending on which ending a player is working toward. When on the genocide route, the protagonist will automatically engage Monster Kid in combat, bringing up the enemy encounter window. If players then choose to continue their extermination of the Underground with an attack, they will find that Undyne intervenes by taking the place of the young monster and allowing herself to be hit with the fatal strike. This juxtaposition of selflessness with the protagonist's murderous intent is another way Fox tries to clarify the notions of good and evil. Through determination alone, Undyne manages to survive and then attacks the player's avatar as the track “Battle Against a True Hero” (Ex. 2.27) begins to play. Opening with solo piano, this work which mixes real-world instruments and synthesized sounds bursts into an intense forte at measure thirteen. And aside from its use of percussion,

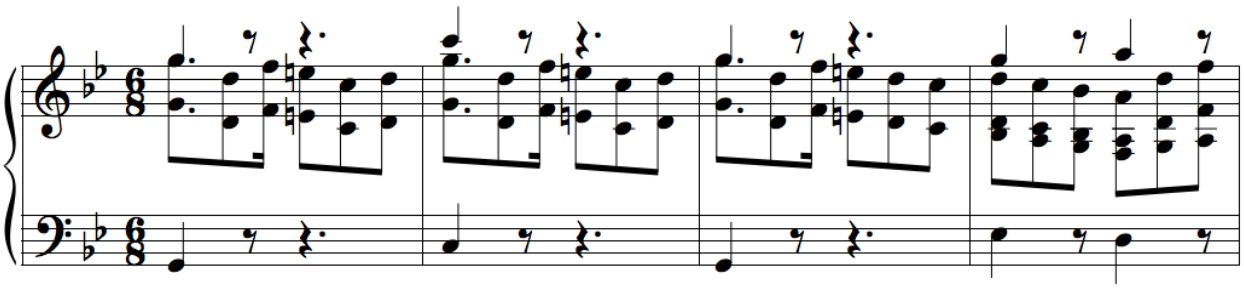
*Example 2.27. Battle Against a True Hero, m. 13-14*



*Example 2.28. Battle with Undyne*




*Example 2.29. Spear of Justice, m. 1-4*



what gives this track its rhythmic ferocity is its blazing fast bassline.

For players on the neutral and true pacifist routes, this confrontation takes place at a different in-game location as opposed to where it occurs in the genocide route. This disparity is a result of Undyne escorting Monster Kid away from the protagonist, thus giving the player an opportunity to move to the next area. Here, Undyne confronts the player (Ex. 2.28) with a

monologue as accompanied by “Spear of Justice” (Ex. 2.29), an explosive track in compound meter. Much like “Battle Against a True Hero”, this work uses a combination of synthesized and orchestral instruments thereby providing it with enough variety to keep the musical texture interesting throughout its nearly two-minute loop cycle. “Spear of Justice” draws upon motives found in “Ruins”, “Undyne”, and “Waterfall”, allowing for it to make connections between the antagonist with which it is associated and previously visited gaming environments.

While it is the first half of *Undertale* that sets up the beginnings of an in-game system of morality, the second half of the game brings it to its logical conclusion. In the following chapter, I will discuss the closing scenes from the title and discuss both the musical and narrative differences between the various routes a player can take. This discussion will also include an analysis of how tonality (or the lack thereof) and changes to the title's soundtrack are used to reinforce *Undertale*'s presentation of good and evil. Moreover, my thesis will conclude with an extensive dissection of the problematic moral implications inherent to the various scenarios in which the protagonist continually finds themself. In doing so, I will bring my arguments to a close and show how Fox's work (though confusing in its presentation of ethics) follows through with its promise of a choice-driven gameplay experience.

## *Chapter Three: Facing the Consequences*

A game about ethics and player choice is only as good as its ending (or endings), a concept which will serve as the topic of discussion in this final chapter. Thus far, we have seen how *Undertale* affords its players a variety of choices which alter the course of its narrative, but now the way in which this title comes to a close will be analyzed. Indeed, Fox's work demonstrates a unique harmony between decision-making and narrative as reflected in its score, and it is this synthesis (which has popularized *Undertale* among its fans) that is further expanded upon in the title's varying conclusions. Players on the true pacifist route are rewarded for their good works while those working toward the genocide ending are punished for their unethical behavior. As for the neutral route, a player on this narrative path is judged according to their actions in regards to how many monsters they have killed. Though it is hinted to players throughout the game as to which route their avatar is currently on, there is one scene in particular which brings the title's theme of morality to the foreground. But before we consider that sequence of events, let us examine the moments of gameplay which directed the player's attention to Fox's system of in-game ethics.

One of the most interesting methods of communicating to players is the use of harmonic stability. As a player begins making progress toward the genocide ending, much of *Undertale*'s soundtrack is intentionally distorted for the purpose of punishing game participants' evil behavior. For example, if a player chooses to destroy each of the limited number of monsters in the title's first area they will begin working their way down the genocide route. In this case, players will know they have completed the violent task once the enemy encounter window appears as accompanied by the message “But nobody came.” This eerie screen no longer makes use of the track “Enemy Approaching”, but instead has “Your Best Friend” playing in the

background at an extremely low playback rate (thus transforming it into a distorted, low-pitched dirge). Since it is afterward utilized in place of the area's harmonically stable background track (“Ruins”), this musical transformation is yet another way the game indicates its ethical boundaries to the player. The same is true in other in-game locations. For players on the genocide route, they will discover that Snowdin has been abandoned and that this same distorted version of “Your Best Friend” is now playing in the background. The player who chooses this narrative pathway is also barred from hearing the track “Shop”, which would otherwise play as the protagonist enters the town's item store. These differences help to reinforce *Undertale*'s ethical values in an attempt to modify a player's behavior.

For the player who has meticulously rid the in-game world of all monsters by the time the protagonist is confronted by Papyrus, they are presented with an entirely different scene. Upon approaching the skeleton, the track “BoneTrousle” (slowed down to the point of tonal ambiguity) plays as the child continues to move closer and closer to the skeleton. Papyrus, now frightened, then declares, “IT FEELS… LIKE YOUR LIFE IS GOING DOWN A DANGEROUS PATH.” Eventually, the child reaches the skeleton thus prompting the enemy encounter window, and the choice between violence and pacifism once again emerges as the game's primary performative aspect. But instead of dealing with this situation as accompanied by the playful rhythms of “BoneTrousle”, players are now met with silence. Indeed, if a player chooses to murder Papyrus they must do so with an absence of background music, making this a particularly chilling scene. This alteration to the title's soundtrack can also be seen in the Waterfall region, in which the background track is modified for the purpose of warning players. Much like in Snowdin Forest, the playback rate for “Waterfall” is greatly reduced for those players on the genocide route in an attempt to reinforce the juxtaposition between good and evil according to *Undertale*'s system of

ethics. This brings us to a scene which plays a paramount role in all three routes in regards to the presentation of morality as it relates to choice-driven gameplay.

The event in question is known as Sans' Judgement, a scenario in which the brother of Papyrus evaluates the protagonist's in-game actions. For players on the true pacifist route, the character praises the child's loving ways and dedication to non-violent practices while the track “The Choice” begins to play. This work is unique in that though it uses familiar motivic materials, it was not created by means of either a modified orchestration or rhythmic transformation. Instead, this track was generated from an altered sound clip taken from a previously used work (“Undertale”) which plays during the protagonist's exploration of the Underground's capital city. And much like a number of earlier tracks that were distorted to reflect a player's faithfulness to the genocide route, the editing techniques used to transform this segment of audio into the track “The Choice” include a dramatic decrease in tempo and the addition of reverberation. What is interesting to note about this work is that it is based on motives from the track “Once Upon a Time”, and that its melodic profiles are stretched over large groups of measures. Furthermore, its atmospheric qualities complement the uplifting nature of Sans' monologue to the protagonist. “The Choice” will also play for players on the neutral route during this scene even if they have killed monsters during their journey. However, the skeleton's words are altered to reflect such circumstances, bringing to light the protagonist's ability to think about and make their own decisions.

Players working toward the genocide ending are greeted with an entirely different experience in terms of both gameplay and music. Instead of being given a speech about morality, they are questioned and threatened. This is followed by a transition to the enemy encounter window (Ex. 3.1) as accompanied by the following monologue:

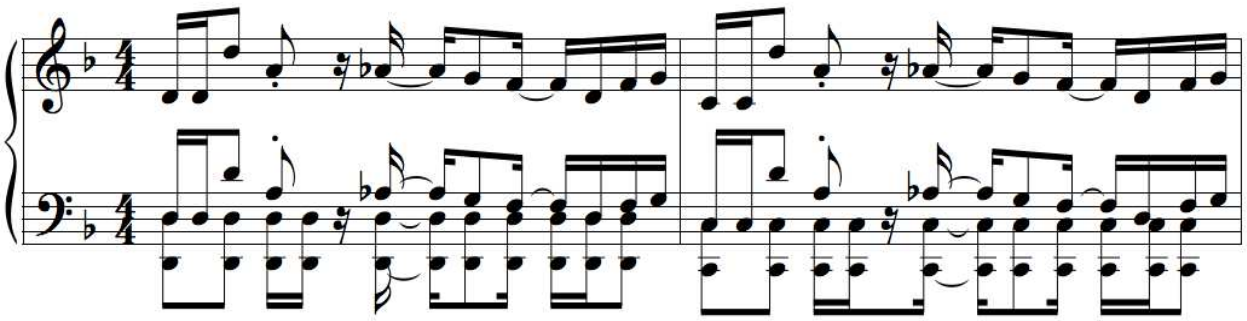
*Example 3.1. Battle with Sans*




it's a beautiful day outside. birds are singing. flowers are blooming… on days like these, kids like you… Should be burning in hell.

The player's avatar is then made to engage Sans in combat as the track "MEGALOVANIA" (Ex. 3.2) begins playing in the background. Opening with a solo synthesizer, this work bursts into a frenzy of notes in its ninth measure with a combination of electronic, orchestral, and rock and roll sounds. And since this is arguably the most challenging battle in the entire game, the track's intensity complements the difficult circumstances a player must face when combating the skeleton. Furthermore, "MEGALOVANIA"'s length of two minutes and thirty-six seconds is appropriate for the confrontation since this is one of the longest fights a player will experience in any of the three narrative routes. The work's persistent bassline gives the track a ferocity

*Example 3.2. MEGALOVANIA, m. 9-10*



throughout its various sections, and its varying instrumentation continually refreshes the sonic atmosphere during the combat scenario. Moreover, this track is unlike its predecessors in that it does not make use of any motives from either previous or subsequent works. Instead, "MEGALOVANIA" is counted among one of the game's tracks which were written before the development of *Undertale* began.

In regards to the title's development of themes, this work acts as the crux of Fox's confusing message about ethics and decision-making in video games. For not only does it appear at the peak of the genocide route's climax, but it has since become one of *Undertale*'s most popular tracks. As of November 2016, "MEGALOVANIA" has been listed as the most individually downloaded work from the game's soundtrack on the iTunes Store, and a brief Google search only further serves to demonstrate its popularity. When searching for "undertale soundtrack", the first search prediction to appear after "undertale soundtrack download" is "undertale soundtrack megalovania". Indeed, to reward players for their murderous ways with one of *Undertale*'s most compelling pieces of music is to turn the idea of punishing immoral behavior on its head. Nevertheless, there are many fans who choose to maintain a more idealistic perspective on the title's sense of morality. These gamers refuse to participate in the genocide route, and instead prefer to watch let's play videos in which YouTube personalities perform the

game's most brutal scenes. And while this might seem like a practical solution for those who desire to abstain from digital violence, some of these fans "criticize Genocide players for their cruelty while also using them as moral sacrifices so 'good' players can watch recordings of the Genocide content, without the Genocide guilt."[1]

After progressing from Sans' Judgement, players can proceed to the throne room ahead of them in order to take part in the next series of events. Here, the protagonist encounters Asgore amidst the sound of chirping birds. For players on the neutral and true pacifist routes, the track "Small Shock" (Ex. 3.3) will begin to play as the anthropomorphic goat is made aware of the human child's presence. This work is made up of only two chords: a C major ninth chord and B minor seventh chord. Though these harmonies were taken from the first measure of "Snowdin Town", they harmonize with a number of other tracks which use the motivic materials from "Snowy". Asgore then greets the player, clearly struggling with conflicting emotions:

> Howdy! How can I… Oh. … I so badly want to say, "would you like a cup of tea?" But… You know how it is. Nice day today, huh? Birds are singing, flowers are blooming… Perfect weather for a game of catch. … You know what we must do. When you are ready, come into the next room.

After delivering his lines, the goat leaves the room to wait for the protagonist in the next area. As for players on the genocide route, this scene plays out quite differently.

*Example 3.3. Small Shock, m. 1-4*

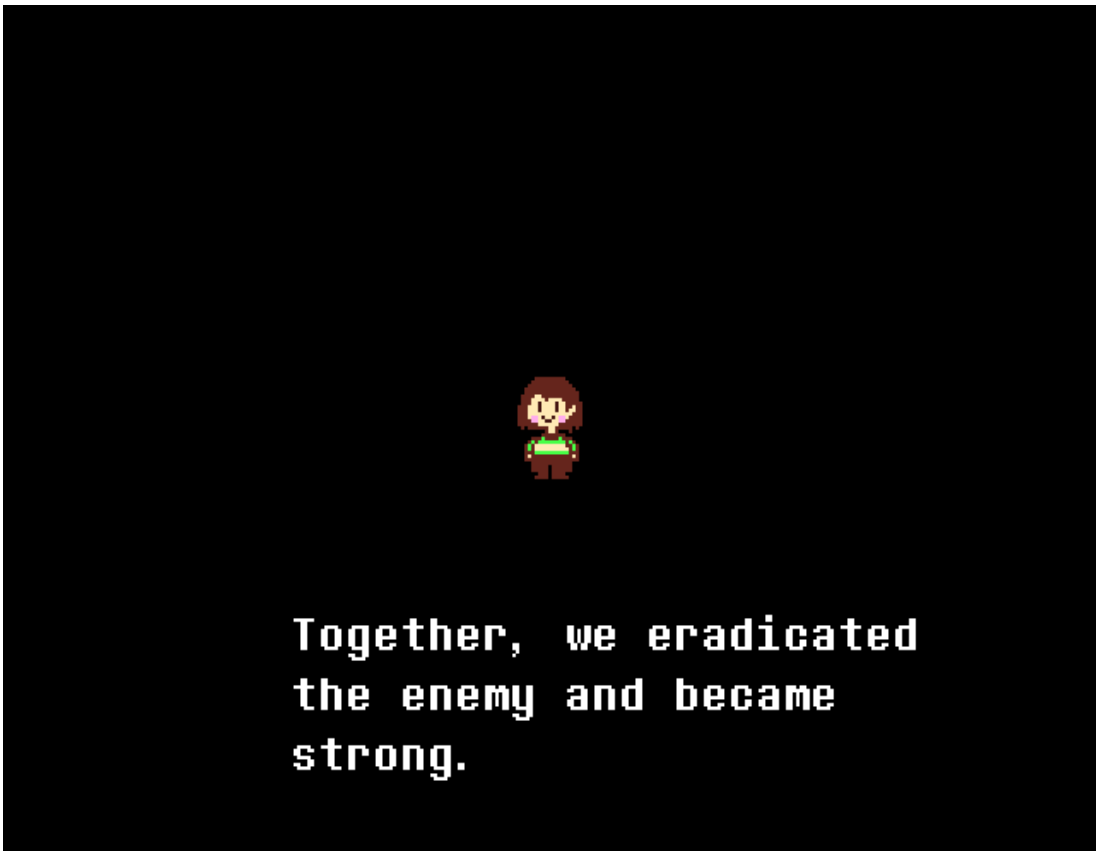


1 "It's OK if You Killed Everyone in Undertale," last modified November 5, 2015, http://blog.plasterbrain.com/2015/11/05/genocide.

Upon entering the throne room, the protagonist is greeted by Asgore who then innocently delivers the following line: “What kind of monster are you… ?” This multifaceted question is meant to serve as a sarcastic jab to a player's character after their completion of the mass extermination required of the genocide route. Such a poignant difference in dialogue is another way in which Fox cleverly reaches out to his audience with his personal message about pacifism, muddled as it may be. The protagonist then initiates the enemy encounter window (thus silencing the throne room's ambient sounds) and, ignoring the goat's plea for a non-violent resolution, attacks with a single, lethal blow. Asgore is subsequently finished off by Flowey who then pleads for his own life, but in this scripted scene it is too late for the player to make such a decision. Instead of being spared, the flower is violently cut into pieces by the now-inhuman child, thereby triggering the final scene of *Undertale*'s genocide route.

After the depiction of Flowey's brutal demise, a single image appears at the center of an otherwise blank screen. What players see here is another human child (that is, Chara) quite similar in appearance to the avatar they have been controlling throughout the title's narrative experience (Ex. 3.4). Chara then introduces themself, and afterwards a high-pitched squealing sound begins to whine in the background. This is followed by a monologue about the decisions that have been made throughout the protagonist's journey as well as the their consequences. Chara, clearly addressing the player and not their avatar, then offers to “erase this pointless world” so that they and the game participant may “move on”. Players are then given the choice between “ERASE” and “DO NOT”, though choosing not to delete the game world they have helped to shape will only cause a disturbing animation of the character approaching the player (after which the game crashes). For those who choose to erase the title's contents, Chara is pleased and comments on how the game participant is a “great partner”. This is followed by a

*Example 3.4. Chara's monologue*



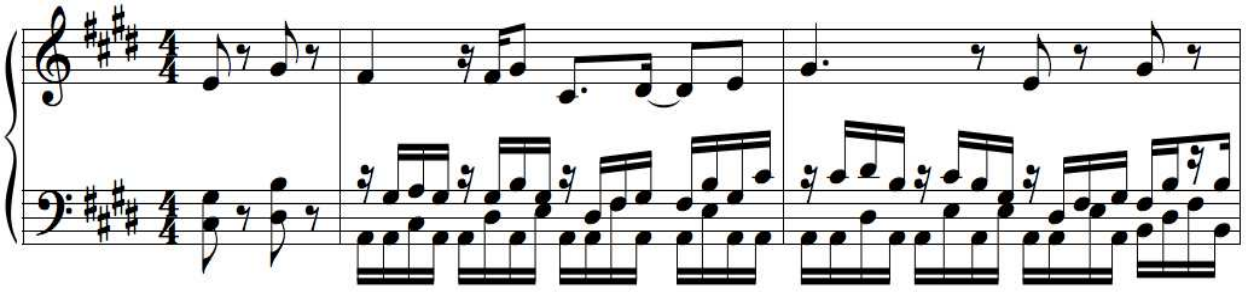

large slicing animation at the center of the screen thus indicating the character's attack on the players themselves, an event which also crashes the game.

Players who then choose to reopen *Undertale* will find that Chara offers to send them back to the world they destroyed in exchange for their soul. However, this transaction is not without consequences as completing the genocide route permanently alters the gameplay experience, thus serving as Fox's final punishment to players who decide to engage in murderous behavior. These changes (discussed later on) are seen when the player chooses to pursue the true pacifist route after the conclusion of their homicidal rampage. Upon its completion, players under such circumstances will find that the closing scenes from *Undertale*'s most peaceful narrative path have been modified to reflect their previously expressed violent intentions. This

permanent change to the gameplay experience demonstrates how the title follows through with its promise of a system of in-game ethics as indicated by alterations to its interactivity and soundtrack. Though the presentation of moral values in *Undertale* may be questionable at best, it is a game which certainly delivers in regards to offering real choices to its players as where "many RPGs still struggle with player morality … because of consequence."[2] Indeed, the permanent consequences of reaching this scene demonstrate how Fox's work forces players to take responsibility for their actions and meditate on their choices.

While the previously described events mark the end of the genocide route, the other two narrative paths continue to unfold from this point. Players on the neutral route next experience a scene in which the protagonist is confronted by Asgore, reluctantly engaging the child in battle as the track "ASGORE" (Ex. 3.5) begins to play. As for the pacifist route, its contents will be discussed later on since it is necessary to first analyze the events leading up to the neutral ending. The work "ASGORE" uses a mixture of synthesized and orchestral sounds to charge its sonic texture with an intensity appropriate for the scenario. From the work's opening measures a persistent bassline continually assaults listeners with a barrage of sixteenth notes, only pausing for cadential moments in this first section. Then, at measure seventeen, there is a change in both

*Example 3.5. ASGORE, m. 1-2*



2 "[Fight] or [Mercy]," last modified October 19, 2015, http://geekandsundry.com/fight-or-mercy-challenging-player-morality-in-undertale.

*Example 3.6A. Heartache, melodic transcription, m. 1-2*



*Example 3.6B. ASGORE, melodic transcription, m. 17-18*




instrumentation and melodic profile. By comparing this new phrase with the opening measures of “Heartache”, it becomes clear that the motivic materials in this section are taken from the earlier work (Ex. 3.6). This reuse of melody is not only used to represent Asgore's past relationship with Toriel, but also the emotional turmoil he feels while fighting the protagonist. Though the goat feels it is his duty to kill the human child and take their soul as a means of freeing his people from their underground imprisonment, he finds no pleasure in the task. Be that as it may, the character's willingness to go through with the act of murdering a child invalidates any arguments about his supposedly benevolent characteristics.

The musical content of “ASGORE” complements the difficulty found in this combat scenario as players on the neutral route must dodge a variety of attacks while simultaneously discerning a peaceful means of resolving the situation. But unlike in other battles, the “MERCY” command is missing from the bottom of the enemy encounter window (Ex. 3.7) due to Asgore having shattered it with his trident, effectively forcing the protagonist to resort to violence. Only upon weakening the goat with repeated blows are players given a choice between sparing and slaying, thus allowing for a continuation of the protagonist's pacifism. However, whether or not the player chooses to kill Asgore is irrelevant since the following sequence takes place regardless

*Example 3.7. Battle with Asgore*



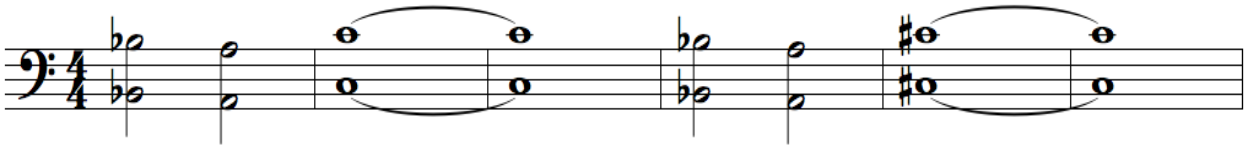

of which decision is made: Flowey makes an unexpected appearance and either destroys both the king's physical form and soul or (if the protagonist has already destroyed Asgore's body) his already exposed, vulnerable soul. The flower then absorbs the other human souls gathered by the king and his monsters, followed by the game suddenly shutting down.

Upon reopening *Undertale*, Flowey thanks the protagonist for either weakening or killing Asgore to the sound of "You Idiot" (Ex. 3.8) playing in the background. Making use of a tonally ambiguous melody, this short, looping work complements the dramatic tension generated by the

*Example 3.8. You Idiot, m. 1-8*



*Example 3.9. Your Best Nightmare, m. 1-6*



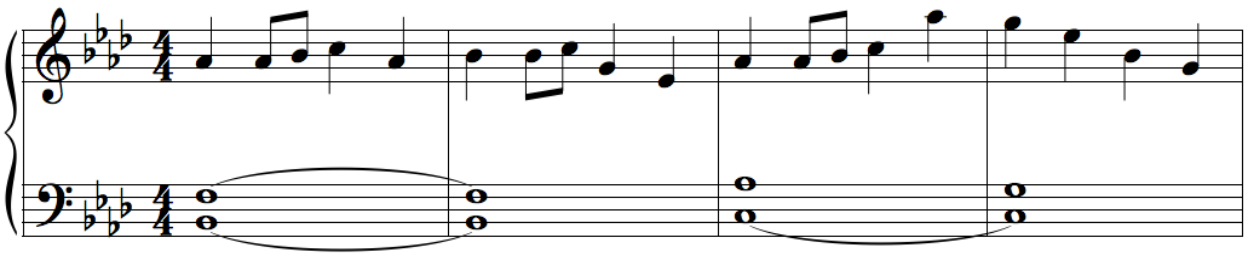
flower's monologue with its buzzing timbre and low-pitched droning. Players then find that the absorption of human souls has transformed the creature into a large, grotesque monster. The monstrosity then approaches the player's avatar with murderous intent as the track “Your Best Nightmare” (Ex. 3.9) begins to play. This work utilizes a slightly altered adaptation of the melody from “You Idiot”, leaping up a minor third instead of the expected major third in its second measure. A similar adjustment to the intervalic contents of the preceding track can be observed in measure five, though what is most noticeable when comparing the two works is their difference in tempi. While “You Idiot” moves along at a slow and sinister pace, this new track presses forward at a blazing 190 beats per minute. As for its instrumentation, “Your Best Nightmare” utilizes the same droning synthesizer as its predecessor as accompanied by a variety of other sounds and electronic percussion. But what really makes this track unique is how it is divided into alternating sections depending on the protagonist's circumstances while battling the nightmarish creature.

After its twelve bar introduction, the track moves into its first division: a thirty second clip which uses the tonally ambiguous motives and buzzing timbre from the work's first six measures as accompanied by an intense breakcore (a genre of electronic music known for its use of highly rhythmic, low-pitched percussion) beat. Afterward, the first of the human souls absorbed by Flowey presents itself, confronting the protagonist. These shifts from one division to another then continue until the protagonist has dealt with all six of the souls contained within

Flowey's form, the musical texture alternating between modified versions of the motivic materials found in “You Idiot” and “Your Best Friend”. In total, there are four variations of the atonal theme, and six variations of the flower's childlike melody. By pairing these musical ideas with contrasting events in the battle, Fox has created an association between tonality and the liberation of souls (an idea which is expounded upon in subsequent scenes). Contrary to this is the composer's use of tonal ambiguity to represent chaos and evil. This concept can be confirmed through an analysis of the genocide route's treatment of distorted music in response to a player's actions, as has already been discussed. But the notion of tonality's positive connotations becomes even more clear in the following segment of gameplay.

After the protagonist has freed the six souls earlier absorbed by Flowey, the player is given the power to fight back against the creature in order to end the battle. It is during this game-changing moment that the track “Finale” (Ex. 3.10) starts to play. Beginning with the soft tones of a solo piano in its introduction, the work bursts into a sudden forte at measure seventeen in which a synth lead takes over as accompanied by a breakcore beat similar (albeit more subdued) to the one found in “Your Best Nightmare”. This tonal stability is meaningful because of how this track marks the turning point of the battle in which the protagonist is finally given the ability to not only save themselves but the captive human souls as well. Not only is the player given a chance to rescue their avatar from death, but they are also given hope for the possibility

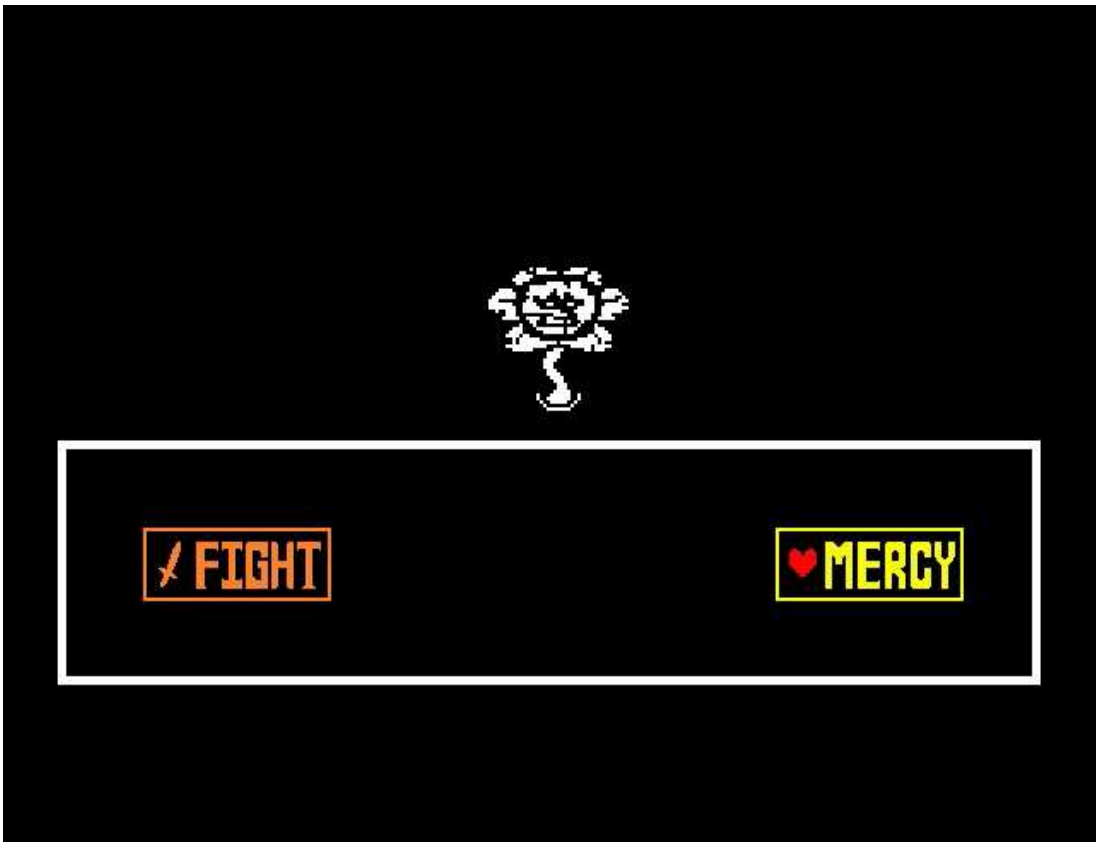
*Example 3.10. Finale, m. 1-4*



of saving other characters as well. Indeed, this positive moment (though tempered by the difficulty of the remaining battle with Flowey) is further reinforced by the uplifting nature of the major mode tonality presented in “Finale”.

Upon the flower's defeat, the protagonist is given the opportunity to either kill or spare the homicidal plant (Ex. 3.11) as complemented by an ambient, wind-like sound effect. And though players are free to choose either option without too much concern for how their neutral playthrough will be affected, this is an important performative moment for those interested in the true pacifist route. If the protagonist decides to have mercy on the flower, he will appear after the title's credits come to a close in order to advise the player on how to achieve a “better” ending depending on what actions they took throughout the game. This harkens to Wolf’s comments on

*Example 3.11. Flowey's defeat*



a game's interactive qualities in that experiential "knowledge, gained from multiple playings, may also be crucial if some of the choices available to the player at a given time are hidden."[3] *Undertale* perfectly captures this idea by encouraging its fans to play through its various endings multiple times, displaying its high level of interactivity.

For players who have already befriended both Papyrus and Undyne (while abstaining from killing any of the subterranean monsters), achieving the true pacifist ending is as simple as returning to the Underground in order to take care of some unfinished business. In short, a player who has fulfilled the above requirements can have their avatar sent back to the point in time just before the scripted confrontation with Asgore for the purpose of discovering the secret behind Flowey's true identity. Triggering the title's final scenes, this series of events takes place in a location known as the True Lab, the track "Here We Are" (Ex. 3.12) playing in the background. In keeping with the theme of tonal ambiguity representing the game's more sinister qualities, this work's lack of functional harmony is consistent with the melancholy atmosphere inspired by the gloomy laboratory. This is significant since the frightening imagery and grim narrative presented in this scene are uncharacteristically dark when compared to the majority of the content found within a neutral or true pacifist playthrough. It is also worth noting that "Here We Are" is one of the few pieces in *Undertale* which uses irregular meter, a rhythmic technique used in the

*Example 3.12. Here We Are, m. 1-4*



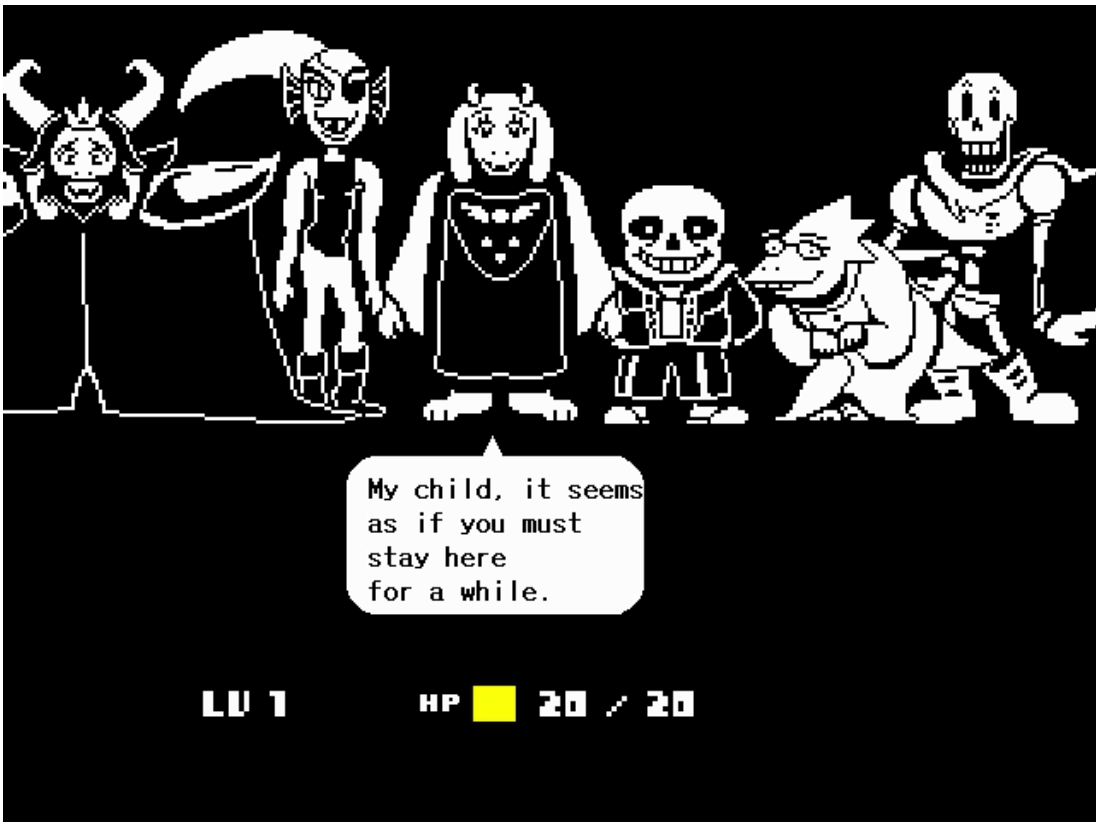
3 Mark J. P. Wolf, "Assessing Interactivity in Video Game Design," *Mechademia* 1 (2006): 82.

composition of “Alphys” (named for the Royal Scientist in charge of the True Lab). Aside from their similar metric structure, the two works share melodic materials as well. As for the piece's instrumentation, it uses synthesized choir sounds as complemented by piano and electronic percussion at a moderate, steady tempo.

In this scene, players discover the origins of Flowey as they unravel the mystery behind his creation. As it turns out, the flower contains the soul of Asgore's dead son as a result of Alphys' experiments. With this piece of information in mind, the rest of the story begins to fall into place. It is no longer a mystery as to why the maniacal creature has played the role of main antagonist throughout the neutral and true pacifist routes. Instead, knowing that Flowey was once the child of Toriel puts the narrative of *Undertale* in a whole new light. Not only does this revelation make the genocide ending seem all the more horrifying, but it motivates players on the true pacifist route to somehow save (or at least put to rest) the soul of this misunderstood character. But as usual, the morality surrounding the protagonist's new objective is questionable at best. Moreover, working toward the true pacifist ending means working toward freeing the Royal Scientist who committed this atrocity in the first place. Ethics aside, the discovery of this background information brings us to the title's final scenes.

After navigating through the True Lab, the protagonist is once again forced to confront Asgore for the purpose of leaving the Underground. But the narrative in this scene unfolds in an unexpected manner due to being framed by a different set of circumstances. Just as the king prepares to engage the child in combat, Toriel swoops in to defend them with her magical powers. She then echoes her very first speaking lines by proclaiming, “What a miserable creature, torturing such a poor, innocent, youth… Do not be afraid, my child.” This singular event dramatically shifts the direction of the title's narrative, having it deviate from the battle

*Example 3.13. Toriel's intervention*




*Example 3.14. Fallen Down (Reprise), m. 33-36*



sequence expected by the player. Instead, what follows is a scene in which Asgore is berated for his actions as the title's main cast joins in for a celebration of friendship (Ex. 3.13). While all of this is taking place, the track “Fallen Down (Reprise)” (Ex. 3.14) proceeds to play softly in the background. This piece begins with a literal quotation from “Fallen Down”, utilizing the same instrumentation and melodic materials from the previous work until the arrival of its thirty-third

*Example 3.15. Fallen Down (Reprise), m. 49-56*



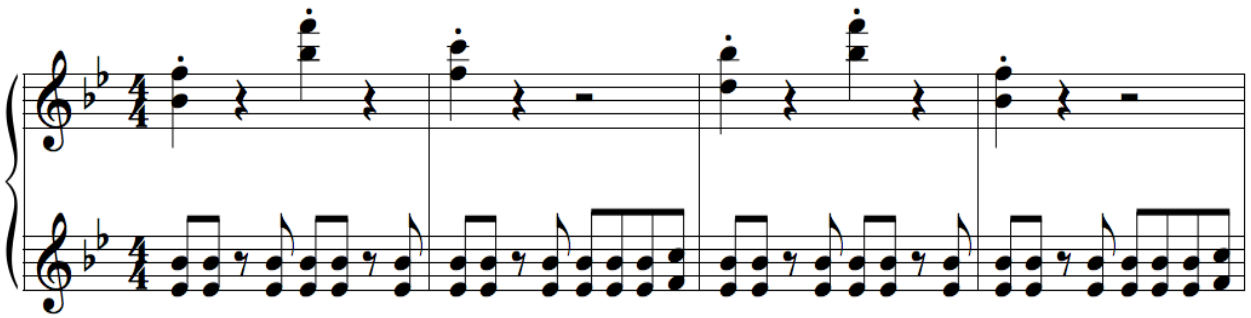
measure. Here, we see a new motivic idea as accompanied by a similar figuration to the one found in the track's earlier section. Later on, Fox makes use of the theme from “Once Upon a Time” to generate a new phrase (Ex. 3.15) in which the accompaniment is slightly altered as a means of refreshing the work's musical texture. In addition to switching to compound meter, this section drops its warm, synthesized orchestration in favor of the simplicity of a solo piano. Subsequently, other instruments are added to the track for the purpose of giving it an emotionally stirring quality. This musical affect lends to the scene's uplifting mood, punctuated by comical dialogue between the title's seemingly goofy characters.

But for the protagonist, a human child who is still trapped beneath the Earth's surface, there is little to be happy about. Indeed, having Toriel stop the confrontation between them and Asgore in order for everyone to live happily ever after in the Underground harkens to the goat's manipulative behavior from earlier in the game. And though she tries to justify her actions by suggesting that the protagonist should not have to choose between murdering Asgore and

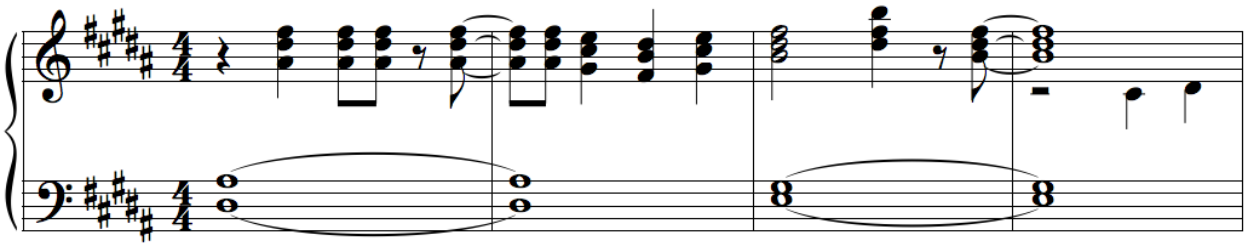
relinquishing their freedom, the implications of her behavior are quite sinister. In actuality, Toriel is making the decision of where the player's avatar will permanently reside for them, effectively restraining the child from exercising their free will. Before the protagonist can get a word in (and before players are offered any kind of choice), the collection of monsters who were only recently trying to either kidnap or murder them consent to the idea through their silence, smiling from ear to ear. Thankfully, this uncomfortable scene (intended to tug on the audience's heartstrings) is interrupted by the violent arrival of Flowey.

Upon binding these characters with his thorny vines, the flower proceeds to absorb every soul in the Underground (except for that of the child) as a means of transforming into his original form, that is, Asriel the goat. The character then engages the protagonist in combat, making for the title's final battle as the track “Hopes and Dreams” (Ex. 3.16) begins playing. Starting out with bright, synthesized strings, this work immediately invokes a positive atmosphere with its major mode tonality, utilizing the melody from “Once Upon a Time”. This music, along with the message first seen in the enemy encounter window which states, “It's the end”, helps indicate to players that they have reached the final scene of the pacifist route. Not only that, but the uplifting quality of this track's opening measures acts as a reward for the peaceful behavior required to access this ending. In measure sixteen, an electric guitar makes its entrance while a lead synth

*Example 3.16. Hopes and Dreams, m. 1-4*



*Example 3.17. Hopes and Dreams, m. 97-100*



takes over the work's melody. Accompanied by a lively rock and roll drum beat, this section continues to press forward with variations on its melodic profile until motivic materials from the track “Your Best Friend” are used as a contrapuntal device approximately one third into the piece's three-minute duration. Obviously, the return of these motives are in reference to the Flowey first introduced to the protagonist, though the overall mood presented by this work is still rather positive. Indeed, the maintenance of its cheerful attributes is possible not only because of the track's rhythm, but also due to Fox's ability to alter his melodies for the purpose of fitting them into different harmonic schemes. Later on, melodic fragments from “Snowdin Town” are utilized (Ex. 3.17), complementing the jubilant atmosphere generated by the piece. This section acts as a reminder to players that they are not fighting for the freedom of their avatar alone, but struggling to free every one of the monsters in the Underground.

In the next scene, the track “Burn in Despair!” starts to play as the protagonist struggles to survive the creature's violent onslaught (Ex. 3.18). Reusing motivic ideas found in “You Idiot”, this work is particularly expressive of the goat's evil nature due in part to its moderately paced, heavy metal drum beat. Moreover, its melodic profile is presented by an electric guitar, utilizing distortion as a way of bringing out the track's negative connotations. Once again, a lack of tonal clarity in this musical context is what highlights the dismal quality of the situation at hand since the child is seemingly given no chance of survival. However, much like the piece's

*Example 3.18. Battle with Asriel*




length of eighteen seconds, the hopelessness of this scene does not continue for a long period of time. Instead, the player learns through narrative text that though they are unable to save their progress at this juncture, they may be able to "SAVE something else." After the message disappears, the work "Hopes and Dreams" returns by taking the place of "Burn in Despair!" as the battle's background music. Simultaneously, the "ACT" option is replaced by a "SAVE" function, allowing for the protagonist to both find and rescue the souls of their newfound "friends". Much like their fight with Photoshop Flowey, the child's showdown with Asriel comes to an end only when each of the souls contained within him are saved.

Upon the character's defeat, Asriel shatters the barrier responsible for sealing the Underground's residents beneath the Earth's surface. He then relinquishes all of the souls he has

*Example 3.19. Reunited, m. 13-16*



absorbed, giving life back to every monster the protagonist has encountered thus far. This is followed by a scene in which the player's avatar wakes up to discover that they are surrounded by Toriel, Sans, Papyrus, Undyne, Alphys, and Asgore. The characters, apparently worried about the child's safety, are then overjoyed to see that they have regained consciousness while the track "Reunited" (Ex. 3.19) plays softly in the background. This work starts with the soothing timbre of an electric piano, presenting a harmonic figuration before any recognizable motives make their entrance. After its introduction, the melodic profile from "Once Upon a Time" is given to an acoustic piano as played above the electric piano's accompaniment. Also utilizing musical materials from "Snowdin Town", "Reunited" makes use of a variety in both its orchestration and rhythmic patterns for the purpose of maintaining its listeners' interest. As the track is playing, the protagonist is free to talk with their underground acquaintances as much as they want before leaving the caverns, bringing the pacifist route to its conclusion.

As a reward for their kindhearted behavior, players are treated to a collection of scenes in which the main cast of *Undertale* is found integrating into human society. Papyrus is seen driving a sports car, Undyne and Alphys are portrayed as relaxing on a seaside boardwalk, and Toriel can be spotted in front of a school building, implying that she now has a job in the field of education. These scenarios unfold to the sound of "Bring It In, Guys!", a piece which uses an

abundance of motives from previously heard tracks. Among the works it references are “Enemy Approaching”, “Nyeh Heh Heh!”, “Snowy”, “Spear of Justice”, “Fallen Down”, and “Once Upon a Time” to name only a few. This upbeat performance is clearly meant to help make the player feel good about their peaceful tactics, strongly indicating Fox's intent for a morally charged gaming experience. This is further expounded upon in a short scene which follows the end of the track “Bring It In, Guys!”, affected by whether or not the protagonist has previously reached the end of the genocide route.

For players who choose to accept Toriel's offer to have their avatar live with her on the surface (a suggestion she makes after Asriel is defeated), they are met with a short sequence in which the child is depicted as sleeping in a bedroom. Players then see the goat quietly enter, leaving a slice of pie in the center of the room. For those who have not completed a genocide route this scene ends after Toriel has left, but this is not the case for players who have previously murdered each of the title's in-game characters. Under such circumstances, the child in the bed is revealed to be Chara who then faces the camera with glowing, red eyes and an evil smile on their face. What follows is the sound of a slowed down version of Flowey's maniacal laughter, therefore spoiling the end of an otherwise happy pacifist playthrough. This set of circumstances is what players refer to as the “soulless pacifist” ending. If the goat's offer is rejected, players are instead rewarded with a photograph in which the main cast is pictured with the protagonist. But for the player who has completed the genocide route, the photograph is significantly altered. Instead of seeing a nice photo of the child and their eclectic group of friends, the protagonist is replaced by Chara, and each of the main characters has a red X over their face. This alteration is used to punish players who have traversed the genocide route, acting as a permanent mark on their ethical record.

## *Reflections and Conclusion*

*Undertale* affords its players a high level of control in terms of their decision making process, thus motivating gamers to play through the entirety of its varying narrative pathways. This freedom can be observed not only in combat scenarios, but also in the title's dialogue and interactive environments. During the gaming experience, players can choose to say and do the things which they most closely identify with, allowing for the prioritization of their individual preferences. And though this level of control is part of what has helped to contribute to the work's popularity, it is the interactive qualities of the game that draw a player in. Each of the decisions made by game participants ultimately affect the narrative experience, therefore highlighting the interactivity inherent to *Undertale*. Because of the way its endings are heavily dependent on choice-driven gameplay, the title offers its players a sense of empowerment in regards to the decisions they are allowed to make. Furthermore, each of these attributes complements the work's performative qualities, from a gamer's ability to make decisions during the course of its narrative to the option of replaying the title in an entirely different manner. Video game performers have taken advantage of these features, entertaining their audiences by exploring the entirety of the work's interactive content. This too has assisted in popularizing *Undertale* among internet users who frequent YouTube (which now hosts an abundance of let's play videos dedicated to the game in question), bringing us to the role music has played in helping Fox's work to make an impact on the gaming community.

In regards to the title's soundtrack, it runs the gamut of both diegetic and non-diegetic music and sounds. An assortment of tracks play in the background, sound effects can be heard in response to a player's inputs, and ambient noise as well as silence are effectively utilized to aid in the presentation of an immersive experience. This technique of utilizing sound in a variety of

ways (inspired by video game titles from numerous genres) is what makes the work's gaming environments seem legitimately dynamic regardless of the visuals used for their creation. And much like in film music, *Undertale*'s use of varying tracks helps players to understand what is happening in its overarching narrative. Without this musical guidance, game participants might not understand the system of in-game ethics Fox has established until later on in the gaming experience. This concept is further expounded upon in *Undertale*'s neutral, true pacifist, and genocide routes in which music is used to either reward or discourage a player's behavior. But unlike the techniques used in movie scores, these changes in soundtrack (used to reflect the protagonist's actions) are a reactive element unique to video games.

By utilizing harmonic stability as a thematic idea, Fox was able to reinforce his narrative on morality and choice. In order to accomplish this, tracks which are clearly tonal in nature were used to indicate ethical behavior amongst the player and in-game characters. While the cast of characters' “good” monsters are given sonically pleasing themes, those with “evil” intentions are represented by a lack of tonality. Papyrus, in spite of trying to kidnap the protagonist, is given a tonal melody because of his supposedly kind-hearted characteristics. Even violent characters like Undyne and Asgore are given tonal tracks due to the purportedly noble intentions with which they operate. On the other hand, Chara is given no theme at all and is instead represented by an unpleasant, high-pitched squeal. Regardless of this confusing system of ethics, the more closely a player comes to reaching the title's true pacifist ending, the more tonal music they are rewarded with. As for players on the genocide route, they are continually met with distorted tracks and tonally ambiguous segments of music as a means of punishing them for their behavior. Clearly, Fox is associating tonality with morality, and atonality with immorality. Although this connection helps to give his work a cohesive musical structure from beginning to end, it is not

without its flaws. Due to the subjective nature of ethics, *Undertale*'s presentation of in-game morality fails to account for certain perspectives, and is questionable at best.

The neutral and true pacifist routes contain more presentations of tonal music than does the genocide route, and while this musical system may seem to validate the composer's views on digital ethics, it is inherently flawed. Its most noticeable fault can be observed in *Undertale*'s pacifist ending, a conclusion known as the "best" and "most moral" way to complete the game among the majority of the title's fanbase. Although the route continually rewards players with positive music and ethically charged monologues meant to encourage their peaceful behavior, it is morally confusing when looking at it from a realistic perspective. At its ending, the pacifist route has Toriel (the manipulative kidnapper) and Asgore (the child murderer) integrating into human society without first receiving an appropriate punishment for their actions. Moreover, the rest of the monsters from the Underground (only previously trying to either abduct or kill the protagonist) are then let loose upon the Earth's surface, free to do as they please. The genocide route, more realistic in its judgment of the human child's decidedly murderous behavior, is less confusing in terms of its ethical treatment of the player. Indeed, its permanent alteration to the game's code gives players a reason to avoid engaging in the unnecessary bloodbath required of the genocide ending. In reality, the most objectively "moral" of the three routes is probably that of the neutral path. This is not only because of the questionable moral values held by the title's main cast, but also because of the player's responsibility to take care of the protagonist. Young and impressionable, the human child should not be forced to bear the burden of bridging the gap between monsters and humanity. Instead, the most "ethical" thing a player can do is to help the child return to their home, freeing themselves from the bullies, kidnappers, and murderers they have met over the course of their underground excursion. Moreover, players are taught that

violence is acceptable if it is used to save another's life. In many ways, this contradictory sentiment is unfair to players who are so strongly encouraged by the game's rules to refrain from all forms of violence. This oft overlooked point muddles the game's purported high ethical standards as reported by critics such as Jason Thibeault who claims that *Undertale* is "a video game that's proven itself more than capable of serving us gamers a love letter, providing us with real moral choices, and encouraging us to buy into the conceits of the game enough that for most people, playing a genocide run is so thoroughly unpalatable to them that they truly feel they're hurting real people."[1]

It has been said that games other than Fox's unique work "are inconsistent about morality across different facets of gameplay", but I propose that *Undertale* is just as guilty of this design flaw.[2] The confrontation between Toriel and the human child is a perfect example of just one of the title's confusing moments, in which a player may not know what the most ethical course of action is. Considering the goat's behavior (and the aggressive music that accompanies the protagonist's battle with her), players could easily draw the conclusion that the supposedly kind-hearted monster is not so kind after all, necessitating a violent escape from her manipulative grasp. For players not on the genocide route who choose to resolve the situation with violence, they will find that the goat is defeated after enough repetitive blows. During her final moments of life, Toriel implores the protagonist to continue their journey until reaching an exit, ending her monologue with the words, "Be good, won't you? My child." This is then immediately followed by a scene in which Flowey says, "I hope you like your choice. After all, it's not as if you can go back and change fate." This poignant line clues players in to the possibility of trying to reset the

1 "Ethical Gamer: Undertale," last modified May 9, 2016, http://the-orbit.net/lousycanuck/2016/05/09/ethical-gamer-undertale.

2 "[Fight] or [Mercy]," last modified October 19, 2015, http://geekandsundry.com/fight-or-mercy-challenging-player-morality-in-undertale.

game and solving the confrontation with Toriel by means other than voluntary manslaughter. In this way, Fox has provided players with enough subtle hints that point to the choice of pacifism as being not only possible but the most ethical decision that can be made as well. On the other hand, for those players who have already met the requirements for the genocide route and intend to continue on that path, Toriel is killed with a single strike from the protagonist. As she is dying, she delivers the following words to the intentionally violent player, "Y… you… really hate me that much? Now I see who I was protecting by keeping you here. Not you… But them!" Whatever it is a player has chosen to do, there is no turning turning back by any means other than restarting the game. Once a player has decided to continue in the narrative path they have chosen by entering the door once blocked by Toriel, the protagonist is unable to undo the actions they have taken while wandering the Ruins.

When it comes to Papyrus, the monster is killed with a single strike for players on the genocide route. On the other hand, players may choose to spare the skeleton, thus returning them to the neutral path for the remainder of the gameplay experience. This continual allowance of opportunities to diverge from the genocide route is how Undertale supposedly "succeeds where a number of games that promise 'moral choice' fail: by presenting the player with true choices, merging those choices into its gameplay and narrative, and being consistent and firm about the consequences of those choices."[3] However, the oversimplification of ethics in this situation does not take the protagonist's perspective into account. Though Papyrus appears to be a harmless simpleton, he continually threatens the protagonist with phrases such as "I WILL … STOP YOU" and "I WILL THEN CAPTURE YOU!" These menacing words are just as disturbing as some of Toriel's monologues, especially in the context of being spoken to a lost child.

3 "[Fight] or [Mercy]."

Furthermore, Undertale rewards players on the genocide route with additional content. Consider the tracks “Battle against a true Hero” and “MEGALOVANIA”. Since the two tracks are only heard in the context of a genocide playthrough, they act as a sort of reward for players who have decided to kill each of the title's in-game characters. This ethical confusion is further expressed in gameplay, as the protagonist has the opportunity to engage in a challenging yet rewarding battles which are otherwise unseen in the neutral and true pacifist routes. Because of these elements, players become motivated to explore the genocide route in order to see this extra (and particularly interesting) content thereby muddling the game's moral standards. This conflict between what a player wants and actually does is what ultimately nullifies *Undertale*'s message regarding love and pacifism, as its players tend to explore the entirety of the title's gameplay experience (whether directly or indirectly) in order to satisfy their morbid curiosity. Having that said, it has been noted by Wolf that a “game’s replayability often depends on its having a good number of options and choices”.[4] This sentiment complements the observation that although *Undertale* is somewhat lacking in sincerity when it comes to its moral underpinnings, it is certainly a title which affords its players a variety in both choice and gameplay experience. For this reason alone it can be said that the title was successfully designed in that it demands to be explored from beginning to end more than once.

It is unquestionable that Fox's abilities as a composer are impressive, making use of a variety of compositional techniques in his work. His reuse of motivic materials from one track to the next give *Undertale* its united musical structure, elevating many of its melodic ideas to leitmotifs representative of characters and complex emotions. And just as operas and musicals utilize music to help audiences understand the progression of their narrative, Fox's soundtrack

4 Mark J. P. Wolf, “Assessing Interactivity in Video Game Design,” *Mechademia* 1 (2006): 80.

acts as a collection of sonic cues which inform players of changes in the protagonist's ethical orientation. Furthermore, the composer's use of tonality and tonal ambiguity to strengthen his system of in-game morality (confusing as it may be) is both impressive and worth studying. It may be valuable for musicologists in the field of ludomusicology to try using a similar approach when studying other choice-driven games. In doing so, scholars can identify trends within the soundtracks of games which deal with ethics.

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